Available online at www.sciencedirect.com

ScienceDirect

Advances in Space Research 76 (2025) 2354–2372

ADVANCES IN SPACE RESEARCH (a COSPAR publication)

www.elsevier.com/locate/asr

Short-term space occupancy and conjunction filter

Ana S. Rivero [a], Claudio Bombardelli [b], Rafael Vazquez [a,*]

[a] Universidad de Sevilla, Escuela Técnica Superior de Ingeniería, Camino de los Descubrimientos s.n., Sevilla 41092, Spain
[b] Space Dynamics Group, Technical University of Madrid, Madrid 28040, Spain

Received 8 January 2025; received in revised form 7 June 2025; accepted 15 June 2025
Available online 18 June 2025

Abstract

Conjunction analysis (CA) for resident space objects (RSOs) is essential for preventing collisions in an increasingly crowded orbital environment and preserving the operational integrity of satellites. A first and fundamental step in the CA process is to estimate the range of altitudes that each object can occupy throughout an operational screening time of, typically, a few days. In this paper, a method is proposed to analytically evaluate such range of altitudes in a zonal problem model and for a time horizon of generic duration thereby generalizing the concept of space occupancy (SO) introduced in a recent work. The proposed method is exploited to construct a new conjunction filter that considerably improves the classical apogee-perigee filter routinely employed in CA. The effectiveness of the new filter is assessed in a low-Earth orbit (LEO) scenario using a high-fidelity perturbation model across a broad spectrum of orbits and conjunction geometries.
© 2025 The Author(s). Published by Elsevier B.V. on behalf of COSPAR. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Keywords: Satellite conjunction screening; Apogee-perigee filter; Space occupancy

1. Introduction

The space environment is becoming ever more crowded, both because of the deployment of large constellations of small to medium-sized satellites by several companies, as well as the growing number of space debris. Over the next decade, more than 20,000 satellites are projected to be placed in orbit, as proposed by approximately two dozen companies (Muelhaupt et al., 2019). Providing perspective for the magnitude of this number, it is worth noting that since the inception of the space age, fewer than 8,100 payloads have been launched into Earth orbit (Muelhaupt et al., 2019). On the other hand, the quantity of small, undetectable orbital debris is increasing progressively and exceeds the number of large, cataloged space objects by several orders of magnitude (Adushkin et al., 2020), further compounding the risk of collisions. Consequently, the probability of a fatal incident resulting in satellite loss is increasing and not negligible (Sgobba and Allahdadi, 2013) and if such an event were to occur, it would not only destroy the two objects involved but also generate a significant amount of debris, which can in turn produce more collisions. The impact between an Iridium satellite and COSMOS 2251, which took place in 2009 is a clear example of such circumstances (Pardini and Anselmo, 2014). To mitigate the risk of collisions, the initial step involves detecting close encounters between satellites of interest and other objects. A close encounter, or critical conjunction, occurs when the expected orbits of two satellites result in a minimum approach distance below a few kilometers. Once a critical conjunction is identified, the threat level is assessed by computing the probability of collision. Finally, if this probability is above a given threshold, a collision avoidance maneuver is performed (Krage, 2020).

Considering the substantial number of Resident Space Objects (RSOs), this process is computationally intensive.

* Corresponding author.
E-mail addresses: asanchez8@us.es (A.S. Rivero), claudio.bombardelli@upm.es (C. Bombardelli), rvazquez1@us.es (R. Vazquez).

https://doi.org/10.1016/j.asr.2025.06.040
0273-1177/© 2025 The Author(s). Published by Elsevier B.V. on behalf of COSPAR.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

The "all on all" conjunction screening problem is seldom addressed operationally due to its inherent difficulties: comparing an entire catalog of more than 20,000 objects against itself leads to screening over 400 million pairs. This challenge is poised to intensify in the future due to increased space traffic and enhanced observation capabilities, which will significantly expand the catalog of objects and, consequently, the number of potential conjunction pairs (Stevenson et al., 2023). Nonetheless, prior research has explored various strategies to mitigate this calculation load, such as filtering processes and parallelization (Escobar et al., 2012; Hall et al., 2009; Healy, 1995; Woodburn et al., 2009). Specifically, applying filters to exclude RSO pairs with negligible collision risk is one of the most commonly used methods to reduce the computational burden in conjunction analyses (Kerr and Sanchez, 2021). Literature proposes several filter types, notably apogee-perigee (AP), orbit path, and time filters. The first two are purely based on the geometry of the orbit pair (Hoots et al., 1984) while the latter takes phasing into account. The AP-filter eliminates pairs with non-overlapping altitude bands, precluding possible collisions. The orbit path filter excludes pairings with a consistently greater distance than the safety threshold. In contrast, the time filter relies on temporal coincidence of orbits which are close. The fact that two orbits are within a threshold distance from each other does not guarantee a conjunction; the satellites traversing these orbits must also be within the specified proximity simultaneously (Algethamie and Armellin, 2020).

The AP-filter stands out for its straightforwardness among the aforementioned screening methods. It selectively excludes pairs of objects whose Earth-centric radii cannot intersect, determined by the minimal and maximal radial values derived from perigee and apogee calculations, respectively. This evaluation necessitates only a single computation per object, specifically using the explicit formulas to determine perigee and apogee radii, which significantly reduces the processing demand. This adjustment allows the "all on all" conjunction analysis to scale linearly with the size of the population, rather than quadratically. Nevertheless, the reliance of this method on a Keplerian framework introduces a notable limitation. Perturbations, chiefly among them gravitational harmonics, lead to variations in orbital elements, subsequently causing alterations in perigee and apogee distances (these effects are approximately two orders of magnitude larger than those caused by the next most relevant perturbations: lunisolar effects (Vallado, 2001)). Thus, introducing an adjustable buffer zone is essential to accommodate these changes. Establishing such a safety threshold, however, is inherently heuristic and demands careful consideration (Alfano and Finkleman, 2014). An undersized buffer may overlook actual conjunction threats, whereas an oversized one could unnecessarily retain numerous non-critical pairings. This situation underscores the screening methods' vulnerability to two types of errors: false positives, where non-threatening pairs are marked for further evaluation, and false negatives, where potential conjunctions are prematurely dismissed.

While the AP-filter remains the first stage of most conjunction filtering techniques, several studies have proposed enhancements to its implementation. For example, Filter I of Casanova et al. (2014) addresses two problems associated with AP version defined by Hoots et al. (1984). Initially, it computes the apogee and perigee for each object using osculating elements, thus addressing the first issue of sensitivity to both short-term and long-term perturbations. The second concern is the risk of false positives during the comparison of apogee and perigee for two objects at a specified time, especially if one is an active satellite executing a maneuver. This is mitigated by utilizing ephemerides tables to determine equinoctial orbital elements, which are then linearly interpolated to estimate these parameters at any given moment. The core strategy of this filter involves shifting from direct apogee and perigee calculations to determining the geocentric global extrema of the distance at a precise time. Woodburn et al. (2009) suggest two additional enhancements to this approach. The initial strategy adds padding to the detection threshold and examines each trajectory at the start and end of the analysis interval, assessing the radial distance variations for each object. This dual-point sampling aims to account for apogee or perigee trends. The alternate strategy, akin to Filter I, opts for selective sampling across all trajectories to ascertain the minimum and maximum radial distances of each one, albeit at a higher computational cost.

A recent study by Bombardelli et al. (2021) introduces the concept of space occupancy (SO) as a means to organize the distribution of Resident Space Objects into non-overlapping orbital shells, thereby minimizing the number of required collision avoidance maneuvers. The work was focused on exploring potential solutions for optimizing RSO distribution over the long term (a few months or longer), considering SO on a time scale sufficient for most LEO objects to complete a full rotation of the line of apsides. That concept, denoted as "long-term SO" in the present work, was shown to be closely linked to the geometry of frozen orbits and the notion of proper eccentricity, first introduced by Cook (1966).

For conjunction screening, which operates within significantly shorter time horizons (days) compared to the precession period of the line of apsides (typically, months), the established SO theory, as detailed in Bombardelli et al. (2021), proves to be overly conservative for most scenarios. By addressing this gap, the central contribution of the current work is the introduction and development of a "short-term SO" theory, which offers a tailored methodology for accurately and swiftly estimating the altitude range within which a satellite operates over specific, short timeframes, achieved through solving a quartic equation. This refined approach supports the creation of a novel conjunction filter, termed the Space Occupancy conjunction filter (SO-filter), which surpasses the traditional AP one by

precisely accounting for the influence of zonal harmonics on a trajectory of the RSO. The SO-filter, based on a zonal-perturbed two-body problem, emerges as a superior alternative for conjunction analysis.

The structure of the article is as follows. First, Section 2 revisits the SO theory and lays the theoretical groundwork for short-term SO, essential for the operational advancements introduced later. The results are validated using a high-fidelity propagator. Next, Section 3 describes the development and analytical implementation of several filters based on the short-term SO model. Their performance is then compared to that of the classical AP-filter, focusing on reliability and effectiveness. Finally, the article concludes with some final remarks in Section 4.

2. Space occupancy theory

This section starts by revisiting the foundational concepts of Space Occupancy (Section 2.1), referred to here as long-term SO due to its focus on long time intervals. In contrast, the primary contribution of this work is centered on short-term SO, which is more useful for operational tasks such as conjunction analysis. The novel short-term SO theory is introduced in Section 2.2, which includes a comprehensive numerical validation using a high-fidelity propagator.

2.1. Long-term space occupancy

SO was introduced in Bombardelli et al. (2021) as the complete geometric domain that a satellite occupies as it traverses its nominal orbit, under the influence of environmental perturbations, during a defined interval. This concept parallels the reachability domain (Holzinger and Scheeres, 2009), used in operations such as spacecraft rendezvous or formation flying (Sanchez et al., 2021), where thrust accelerations define accessible spatial regions. Yet, in contrast, SO is uniquely governed by environmental perturbations. Notably, when the interval under study coincides with or exceeds the J_2-induced apsides precession period, SO becomes intimately connected with the geometry of frozen orbits and the notion of proper eccentricity, concepts originally introduced by Cook (1966). These elements provide analytical insights into RSO orbital dynamics, enhancing our comprehension of SO. The contributions of Bombardelli et al. (2021) and some notions from the work of Cook (1966) on frozen orbit theory are succinctly reviewed below for the reader's convenience.

Consider the Earth's radius, $R_\oplus$, as the reference length unit and $1/n_\oplus$ as the reference time, with $n_\oplus$ the mean motion of a circular orbit of radius $R_\oplus$. The dimensionless time, denoted by τ, is then defined as the ratio of the elapsed time to the reference time scale $1/n_\oplus$. Let $\hat{e}$, $\hat{\omega}$, $\hat{a}$, and $\hat{i}$ represent the mean values—obtained by averaging these parameters over a full orbit—of the eccentricity, argument of periapsis, semimajor axis, and inclination, respectively (Vallado, 2001).

The foundational work of Cook (1966) describes the behavior of the mean eccentricity vector of an orbit influenced by J_2 and an arbitrary sequence of odd zonal harmonics, with components (ξ, η) in the nodal frame given by

$$\xi(\tau) = \hat{e}(\tau)\cos\hat{\omega}(\tau), \qquad \eta(\tau) = \hat{e}(\tau)\sin\hat{\omega}(\tau). \tag{1}$$

For orbits of small eccentricity, these vectors trace a circular path ("Cook's circle") defined by

$$\xi(\tau) = e_p\cos\beta(\tau), \tag{2}$$

$$\eta(\tau) = e_p\sin\beta(\tau) + e_f, \tag{3}$$

where e_f, e_p, and $\beta(\tau)$ symbolize the frozen eccentricity, proper eccentricity, and rotation angle, respectively, as illustrated in Fig. 1. Notably, $(0, e_f)$ represents the center of Cook's circle with e_p as its radius, indicating that e_f can be either positive or negative, while e_p remains nonnegative. Cook's theory meticulously provides the analytical expressions for these parameters (Cook, 1966; Bombardelli et al., 2021):

$$e_f = k^{-1}\hat{a}^{-3/2}\sum_{n=1}^{N}\frac{J_{2n+1}}{\hat{a}^{2n+1}}\frac{n}{(2n+1)(n+1)}P^1_{2n+1}(0)P^1_{2n+1}(\cos\hat{i}), \tag{4}$$

$$e_p = \sqrt{\left(\hat{e}_0\sin\hat{\omega}_0 - e_f\right)^2 + \hat{e}_0^2\cos^2\hat{\omega}_0}, \tag{5}$$

$$\sin\alpha = \frac{\hat{e}_0\sin\hat{\omega}_0 - e_f}{e_p}, \tag{6}$$

$$\cos\alpha = \frac{\hat{e}_0\cos\hat{\omega}_0}{e_p}, \tag{7}$$

$$\beta(\tau) = k\tau + \alpha, \tag{8}$$

$$k = \frac{3J_2}{\hat{a}^{7/2}}\left(1 - \frac{5}{4}\sin^2\hat{i}\right), \tag{9}$$

where P^1_n denotes the associated Legendre function of the first order and degree n, and J_{2n+1} are the odd zonal harmonic coefficients, see e.g. Vallado (2001). The term k, as defined in Eq. (9), modulates the rotation of the eccentricity vector. Initial conditions $\hat{e}_0$ and $\hat{\omega}_0$ represent the mean eccentricity and argument of periapsis at $\tau = 0$ and α corresponds to the initial condition of β.

From these expressions, the mean eccentricity and argument of periapsis can be readily computed as:

$$\hat{e}(\tau) = \sqrt{e_f^2 + e_p^2 + 2e_f e_p\sin\beta(\tau)}, \tag{10}$$

$$\hat{\omega}(\tau) = \mathrm{atan2}(\eta(\tau), \xi(\tau)). \tag{11}$$

Cook's theory lays the groundwork for the comprehension of frozen orbits, which result from a combination of orbital elements such that the proper eccentricity e_p in Eq. (4) becomes zero, thus making the mean eccentricity of the orbit constant and equal to $|e_f|$. A significant contribution of SO theory is its application of frozen orbits as a framework to ascertain the long-term radial behavior of generic

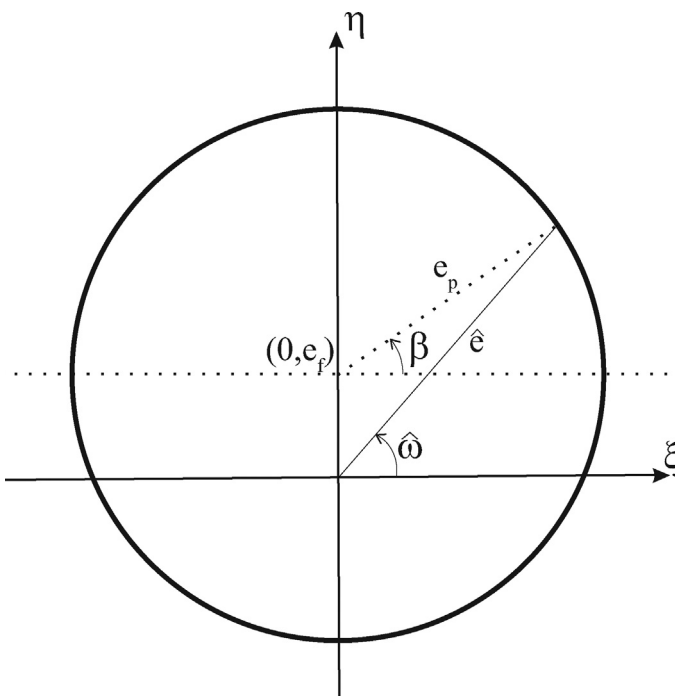

Fig. 1. Motion of the eccentricity vector in the (ξ, η) plane.

orbits with small eccentricities. This involves first determining the maximum and minimum radii of frozen orbits. Considering that the values of $\hat{a}$ and $\hat{i}$ remain constant, a polar equation for the radius of a frozen orbit, denoted as r_f, as a function of the mean argument of latitude $\hat{\theta}$, expanded to first-order eccentricity, is derived in Bombardelli et al. (2021):

$$r_f\left(\hat{\theta}\right) = \hat{a}\left(1 - e_f \sin\hat{\theta}\right) + \frac{J_2}{4\hat{a}}\left[\left(9 + \cos 2\hat{\theta}\right)\sin^2\hat{i} - 6\right]. \tag{12}$$

The maximum and minimum values of the radius[1] as given by Eq. (12), denoted as r_N (minimum radius) and r_S (maximum radius), are described by[2]:

$$r_N \simeq \widehat{a}\left(1 - |e_f|\right) + \frac{J_2\left(4\sin^2\widehat{i} - 3\right)}{2\widehat{a}}, \tag{13}$$

$$r_S \simeq \hat{a}\left(1 + |e_f|\right) + \frac{J_2\left(4\sin^2\hat{i} - 3\right)}{2\hat{a}}. \tag{14}$$

Ordinarily, the minimum radius is attained at the orbital "North" and the maximum at "South" for Earth orbits, due to the specific values of Earth's J_2 and odd zonal harmonics making e_f in Eq. (4) positive[3]. These extrema can switch roles if e_f is negative, a scenario that can only arise (for Earth) in orbits nearing the critical inclination (63.4 or 116.6 deg). Note that in those cases the offset Δ, defined by Eq. (15) and depicted in Fig. 2, which is typically negative (southward) can become zero or positive (northward).

$$\Delta = r_N - r_S = -2\hat{a}e_f. \tag{15}$$

Utilizing frozen orbits, SO theory demonstrated that the radial bounds for the long-term SO of *any* small eccentricity orbit influenced by J_2 and arbitrarily many odd zonal harmonics can be calculated as follows based on the corresponding frozen orbit with the same values of $\hat{a}$ and $\hat{i}$, incorporating an additional quantity, the *space occupancy radius* (SOR):

$$r_{min}^{long} = r_N - \frac{\mathrm{SOR}}{2} = r_N - \widehat{a}e_p, \tag{16}$$

$$r_{max}^{long} = r_S + \frac{\mathrm{SOR}}{2} = r_S + \widehat{a}e_p, \tag{17}$$

where r_{min}^{long} and r_{max}^{long} represent, respectively, the long-term minimum and maximum radial bounds within which the space occupancy of an orbit is confined over an extended period. The SOR, as illustrated in Fig. 2, quantifies the range between the maximum and minimum orbital radii, centered around the corresponding frozen orbit. It corresponds to the difference between the maximum 'mean' apoapsis and the minimum 'mean' periapsis, which can be expressed as:

$$\mathrm{SOR} = (\Delta r)_{\max} \approx a(e_{\max} - e_{\min}). \tag{18}$$

According to Cook's solution, the maximum value of eccentricity corresponds to the sum of e_f and e_p, while the minimum value is given by the difference between e_p and e_f (see Fig. 1). Thus, the SOR is expressed as $\mathrm{SOR} = 2ae_p$.

To illustrate the SO radial bounds, Fig. 3 compares the calculated r_{max}^{long} and r_{min}^{long} values with the mean apogee and perigee for a typical LEO orbit. It also displays the actual radius, computed with a propagator, alongside the osculating perigee and apogee changes over time. This comparison highlights that the orbital radius consistently remains within the bounds predicted by SO theory, unlike the mean apogee/perigee, which may not provide precise bounds. The depicted orbit is characterized by its initial osculating elements: semimajor axis $a = 7,136.6$ km, eccentricity $e = 0.0095$, inclination $i = 1.2723$ rad, argument of perigee $\omega = 1.004$ rad, right ascension of the ascending node $\Omega = 2.0246$ rad, mean anomaly $M = 1.8230$ rad. This

[1] It is crucial to differentiate between maximum and minimum radius of the SO and apogee or perigee concepts. The former are global, potentially conservative bounds for occupied orbital space, whereas the latter, can refer to mean or osculating values. Osculating values change continuously, and mean values may not provide precise bounds, as demonstrated in subsequent sections.

[2] In Bombardelli et al. (2021), the consideration of harmonics up to order 3 in the frozen eccentricity computation results in e_f always being positive, leading to the exclusion of cases near the critical inclination, where e_f could be negative. To address these cases in Eq. (13) and (14), the absolute value has been incorporated.

[3] Here,"North" and "South" denote the northernmost and southernmost positions of the orbit, respectively, achieved when the argument of latitude reaches $\pi/2$ or 90 degrees (respectively $3\pi/2$ or 270 degrees)

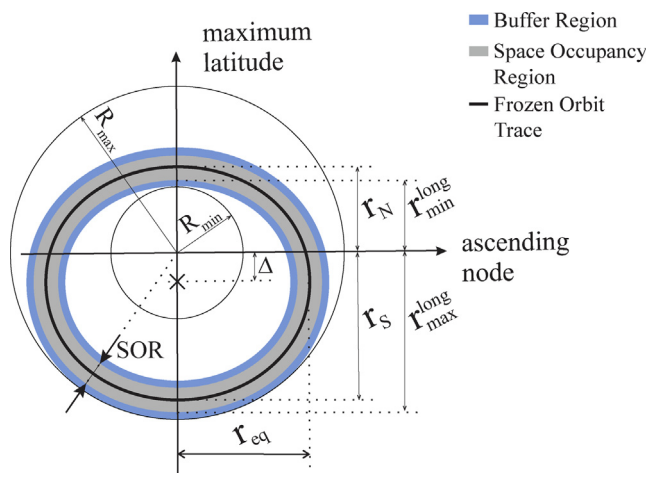

Fig. 2. Space Occupancy Geometry.

figure is generated using the high-fidelity propagator described in Section 2.2.3.

2.2. Short-term space occupancy

When the space occupancy radius needs to be evaluated for time intervals shorter than the line of apsides precession period, as for instance would be the case for conjunction analysis, the long-term SO bounds determined by Eqs. (16) and (17) may prove overly conservative. This motivates a refined space occupancy framework to accurately capture the evolution of the radial bounds of a generic orbit over such shorter timeframes. For this purpose, a polar equation for non-frozen orbits is obtained as a means to provide lower and upper radial bounds over a selected time interval by solving a simple quartic equation. The analytically computed bounds are validated in Section 2.2.3 using a high-fidelity propagator.

2.2.1. A polar equation for non-frozen orbits

The approximate evolution of the radius for a non-frozen orbit with small eccentricity, using a first-order eccentricity expansion, is derived from reference Bombardelli et al. (2021),

$$r \approx \hat{a}\left(1-\hat{e}\cos\left(\hat{\theta}-\hat{\omega}\right)\right)+\frac{J_2}{4\hat{a}}\left[\left(9+\cos\left(2\hat{\theta}\right)\right)\sin^2\hat{i}-6\right]. \tag{19}$$

In order to determine all local maxima and minima of Eq. (19), its time derivative is computed and set to zero:

$$\frac{dr(\tau)}{d\tau}=\frac{\partial r}{\partial \hat{e}}\frac{d\hat{e}}{d\tau}+\frac{\partial r}{\partial \hat{\omega}}\frac{d\hat{\omega}}{d\tau}+\frac{\partial r}{\partial \widehat{M}}\frac{d\widehat{M}}{d\tau}=0, \tag{20}$$

leading to:

$$-\hat{a}\frac{d\hat{e}}{d\tau}\cos\widehat{M}+\hat{a}\hat{e}\sin\widehat{M}\frac{d\widehat{M}}{d\tau} -\frac{J_2\sin^2\hat{i}}{2\hat{a}}\sin\left(2\widehat{M}+2\hat{\omega}\right)\left(\frac{d\widehat{M}}{d\tau}+\frac{d\hat{\omega}}{d\tau}\right)=0, \tag{21}$$

where $\widehat{M}$ refers to the "mean" mean anomaly.

By differentiating Eqs. (2)–(3) and (10)–(11), the time derivatives of $\hat{\omega}$ and $\hat{e}$ are:

$$\frac{d\hat{\omega}}{d\tau}\simeq\frac{1}{\hat{e}^2}\left(\frac{d\eta}{d\tau}\xi-\eta\frac{d\xi}{d\tau}\right), \tag{22}$$

$$\frac{d\xi}{d\tau}=-e_p k\sin\beta, \tag{23}$$

$$\frac{d\eta}{d\tau}=e_p k\cos\beta, \tag{24}$$

$$\frac{d\hat{e}}{d\tau}\simeq\frac{1}{2\hat{e}}\left(2\xi\frac{d\xi}{d\tau}+2\eta\frac{d\eta}{d\tau}\right), \tag{25}$$

and the time derivative of the mean anomaly, retaining only secular terms and neglecting the long-periodic contribution, is (Kozai, 1959):

$$\frac{d\widehat{M}}{d\tau}\simeq\sqrt{\frac{1}{\hat{a}^3}}+\frac{3J_2}{2\hat{a}^{7/2}(1-\hat{e}^2)^{3/2}}\left(1-\frac{3}{2}\sin^2\hat{i}\right). \tag{26}$$

Solving Eq. (21) with Eqs. (22)–(26) provides a highly accurate yet computationally intensive approach for determining short-term space occupancy bounds.

2.2.2. Short-term SO quartic equation

A more efficient method to determine short-term SO bounds can be devised by reformulating Eq. (19). Subse-

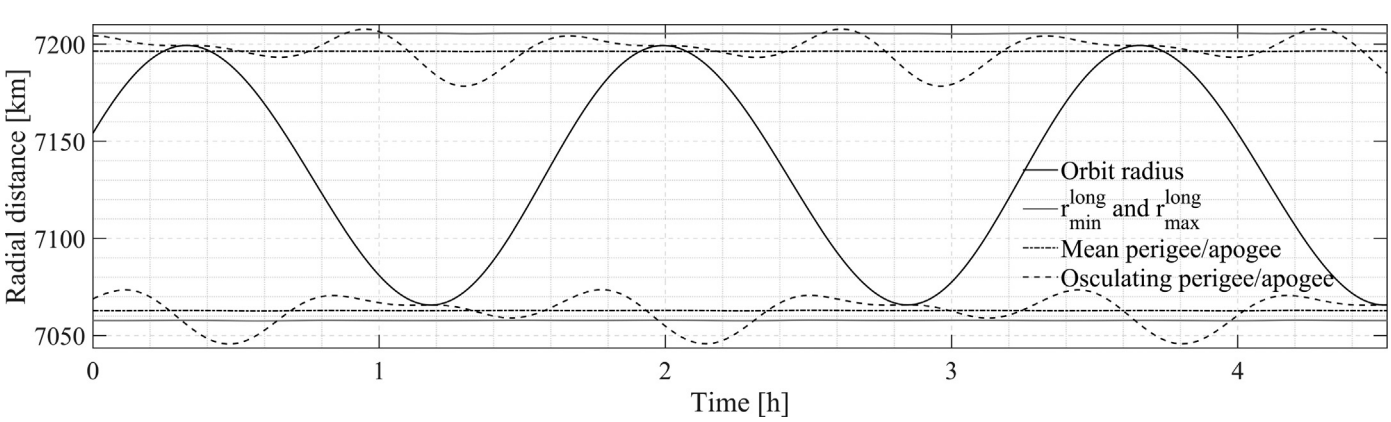

Fig. 3. Comparison of mean and osculating apogee/perigee against long-term SO bounds with actual radius variation for a LEO orbit.

quently, expanding $\cos\left(\hat{\theta}-\hat{\omega}\right)=\cos\hat{\theta}\cos\hat{\omega}+\sin\hat{\theta}\sin\hat{\omega}$ and using Eq. (1), the first term of Eq. (19) can be written as:

$$\begin{aligned}\hat{a}\left(1-\hat{e}\cos\left(\hat{\theta}-\hat{\omega}\right)\right) &= \hat{a}\left(1-\left(\xi\cos\hat{\theta}+\eta\sin\hat{\theta}\right)\right)\\ &= \hat{a}\left(1-e_p\left(\cos\beta\cos\hat{\theta}+\sin\beta\sin\hat{\theta}\right)\right.\\ &\quad\left.-e_f\sin\hat{\theta}\right)\\ &= \hat{a}\left(1-e_p\cos\left(\hat{\theta}-\beta\right)-e_f\sin\hat{\theta}\right),\end{aligned} \tag{27}$$

where Eqs. (2) and (3) have been employed. Incorporating Eq. (27) into Eq. (19) yields an alternative expression for r, directly linking it to $\hat{\theta}$ and β:

$$\begin{aligned}r\left(\hat{\theta},\beta\right) \approx\ & \hat{a}\left(1-e_p\cos\left(\hat{\theta}-\beta\right)-e_f\sin\hat{\theta}\right)\\ &+\frac{J_2}{4\hat{a}}\left[\left(9+\cos\left(2\hat{\theta}\right)\right)\sin^2\hat{i}-6\right].\end{aligned} \tag{28}$$

Given the slow, long-term movement of $\beta=k\tau+\alpha$ and the more rapid changes in the argument of latitude $\hat{\theta}$, it becomes evident that $\hat{\theta}$ crosses its entire range $[0,2\pi)$ significantly faster than β[4]. This observation justifies treating β and $\hat{\theta}$ as independent variables for the purposes of short-term space occupancy thereby converting the time-domain analysis in Section 2.2.1 into a simpler multivariable optimization problem. As r represents a continuous, differentiable function in both β and $\hat{\theta}$, basic calculus principles dictate that its extremal values must occur at either critical (zero-derivative) points or the domain boundaries.

The following is only focused on the problem of finding the maximum of r, noting that a similar procedure should be followed to address finding the minimum of r.

In the long-term SO problem, both $\hat{\theta}$ and β are periodic and therefore the global maximum is always found at critical points as:

$$r_{max}^{long}=r\left(\hat{\theta}_{max}^{*},\beta_{max}^{*}\right).$$

The computation of the critical point giving rise to the global maximum $\left(\hat{\theta}_{max}^{*},\beta_{max}^{*}\right)$, as well as other critical point that correspond to relative maxima $\left(\hat{\theta}_{Lmax}^{*},\beta_{Lmax}^{*}\right)$, as a function of the mean orbital elements is deferred to Appendix A for clarity and conciseness.

Conversely, when addressing the short-term SO problem, and denoting with τ_f the final time, the variable $\hat{\theta}$ crosses its full range $[0,2\pi)$ while β , illustrated in Fig. 1, spans a shorter arc $\beta\in[\beta_0,\beta_1]$ with:

$$\beta_0=\alpha,\qquad \beta_1=\alpha+k\tau_f,$$

where α is defined by Eqs. (6) and (7).

Three possible scenarios arise. If the interval $[\beta_0,\beta_1]$ contains the global maximum value β_{max}^{*} then the solutions for the short-term and long-term SO problem coincide. If the interval contains a local maximum value β_{Lmax}^{*} the radius becomes maximum either at this value or at one of the interval endpoints. Otherwise, the maximum corresponds to one of the interval endpoints. Therefore, the short-term SO maximum can be computed as:

$$r_{max}^{short}\begin{cases} r_{max}^{long} & \text{if} \quad \beta_{max}^{*}\in[\beta_0,\beta_1]\\ \max\left\{r\left(\beta_{Lmax}^{*},\hat{\theta}_{Lmax}^{*}\right),r_0^{max},r_1^{max}\right\} & \text{if} \quad \beta_{Lmax}^{*}\in[\beta_0,\beta_1]\ \wedge\ \beta_{max}^{*}\notin[\beta_0,\beta_1]\\ \max\left\{r_0^{max},r_1^{max}\right\} & \text{otherwise}\end{cases} \tag{29}$$

where:

$$r_n^{max}=\max\left\{r_n\left(\hat{\theta}\right),\hat{\theta}=0..2\pi\right\}\qquad n=0,1,$$

with

$$\begin{aligned}r_n\left(\hat{\theta}\right) &= r_n\left(\hat{\theta},\beta_n\right)\\ &\approx \hat{a}\left(1-e_p\cos\left(\hat{\theta}-\beta_n\right)-e_f\sin\hat{\theta}\right)\\ &\quad+\frac{J_2}{4\hat{a}}\left[\left(9+\cos\left(2\hat{\theta}\right)\right)\sin^2\hat{i}-6\right].\end{aligned} \tag{30}$$

The value of $\hat{\theta}$, denoted with $\hat{\theta}_n^{*}$, that maximizes $r_n\left(\hat{\theta}\right)$, can be obtained by computing the derivative of Eq. (30) and setting it to zero. In this fashion:

$$r_n^{max}=r_n\left(\hat{\theta}_n^{*}\right),$$

where:

[4] The ratio of their periods varies across the dataset: in the orbit with the smallest difference, $\hat{\theta}$ is 354 times faster than β, while in the orbit with the largest difference, the ratio increases to 1.8967×10^8. On average, $\hat{\theta}$ is 29,212 times faster than β.

$$0 = \frac{dr_n}{d\hat{\theta}}\left(\hat{\theta}_n^*\right) = \hat{a}\left(e_p \sin\left(\hat{\theta}_n^* - \beta_n\right) - e_f \cos\hat{\theta}_n^*\right) - \frac{J_2 k}{2\hat{a}} \sin\left(2\hat{\theta}_n^*\right). \tag{31}$$

Eq. (31) can be reduced to a polynomial. Employing the classical substitution $x_n = \tan\left(\hat{\theta}_n^*/2\right)$, which establishes a bijective relationship between $\mathbb{R}$ and $(-\pi, \pi)$, leads to:

$$0 = \hat{a}\left(e_p \frac{2x_n}{1+x_n^2}\cos(\beta_i) - e_p \frac{1-x_n^2}{1+x_n^2}\sin(\beta_i) - e_f \frac{1-x_n^2}{1+x_n^2}\right) - \frac{J_2 k}{\hat{a}} \frac{2x_n\left(1-x_n^2\right)}{\left(1+x_n^2\right)^2}. \tag{32}$$

Cross-multiplying Eq. (32) by $(1+x^2)^2$ results in the quartic polynomial:

$$0 = -\left(\hat{a}e_p \sin\beta_n + \hat{a}e_f\right) + 2x_n\left(\hat{a}e_p \cos\beta_n - \frac{J_2 \sin^2\hat{i}}{\hat{a}}\right) + 2x_n^3\left(\hat{a}e_p \cos\beta_n + \frac{J_2 \sin^2\hat{i}}{\hat{a}}\right) + x_n^4\left(\hat{a}e_p \sin\beta_n + \hat{a}e_f\right), \tag{33}$$

or, in compact form:

$$x_n^4 + Px_n^3 + Qx_n - 1 = 0, \tag{34}$$

with:

$$P = \frac{2\hat{a}^2\hat{e}_n \cos\hat{\omega}_n + 2J_2 \sin^2\hat{i}}{\hat{a}^2\hat{e}_n \sin\hat{\omega}_n}, \quad Q = \frac{2\hat{a}^2\hat{e}_n \cos\hat{\omega}_n - 2J_2 \sin^2\hat{i}}{\hat{a}^2\hat{e}_n \sin\hat{\omega}_n},$$

where the following identities have been employed:

$$e_p \cos\beta_n = \xi_n = \hat{e}_n \cos\hat{\omega}_n = \hat{e}(\beta_n)\cos\hat{\omega}(\beta_n),$$
$$e_p \sin\beta_n + e_f = \eta_n = \hat{e}(\beta_n)\sin\hat{\omega}(\beta_n).$$

Eq. (34) can have up to four real solutions, and can be solved analytically, via Ferrari's method (Strong, 1859), or through efficient numerical strategies, such as Newton's (Polyak, 2007) or Halley's methods (Bateman, 1938).

The consideration of a potential root at infinity requires examining the limit of the polynomial in Eq. (32) when the denominators of both P and Q become zero. This scenario arises exclusively if $\sin\hat{\omega}_i = 0$, implying solutions at $x = 0$ and $x = \infty$, corresponding to $\hat{\theta}_n^* = 0$ and $\hat{\theta}_n^* = \pi$, respectively.

Finally, both the radius and the second derivative (used for discerning the nature of the critical points) can be expressed directly in terms of x as:

$$r_n = \hat{a}\left(1 - \hat{e}_n \frac{(1-x^2)\cos\hat{\omega}_n - (2x)\sin\hat{\omega}_n}{1+x^2}\right) + \frac{J_2}{4\hat{a}}\left[\left(9 + \frac{1-6x^2+x^4}{(1+x^2)^2}\right)k - 6\right], \tag{35}$$

$$\frac{\partial^2 r_n}{\partial\hat{\theta}^2} = \hat{a}\hat{e}_n \frac{(1-x^2)\cos\hat{\omega}_n - (2x)\sin\hat{\omega}_n}{1+x^2} - \frac{J_2 k}{\hat{a}} \frac{1-6x^2+x^4}{(1+x^2)^2}. \tag{36}$$

This approach significantly enhances the efficiency of the short-term SO bounds computation.

2.2.3. Short-term SO validation

The analytical short-term SO model developed in Section 2.2 has been tested and validated against a high-fidelity numerical propagation model[5]. The latter includes a 23×23 geopotential model[6] along with lunisolar third-body perturbations, and accounts for Earth's geoid precession and nutation. The Earth orientation, the values of the harmonic coefficients, as well as the position of the Sun and Moon are obtained from the corresponding SPICE kernels. Note that the analytical short-term SO model is constructed on a zonal problem with the Earth polar axis aligned with the z axis of the inertial frame at the current epoch.

Concerning the applicability limits of the proposed theory, it is important to underline that orbits with eccentricities higher than 0.1 and apogee radii exceeding 40,000 km have been left out. Including these orbits would violate the simplifying assumptions at the basis of Cook's theory, which is a first order theory in eccentricity and neglects third body perturbations, and may result in unacceptable errors. When analyzing the space occupancy of these more eccentric and/or near-GEO orbits a more refined numerical analysis is recommended. Fortunately, the great majority of RSOs are characterized by orbits that are well within the limits of validity of this theory.

Additionally, in this study, a 5-day time window is chosen for the analysis, following standard conjunction assessment practices, where prediction intervals typically range between 3 and 7 days (Krag et al., 2011; Yaglioglu, 2012; Sánchez-Ortiz et al., 2015). While this choice is not a strict limitation of the model, the error associated with the bounding radius increases over time due to unmodeled perturbations such as atmospheric drag or third-body effects. Since LEO conjunction analyses tend to use shorter intervals, the 5-day window represents a reasonable balance, ensuring a practical trade-off between accuracy and applicability for this initial step.

Fig. 4 compares the numerical propagated orbital radius (r) with the radial bounds obtained with the short-term SO model over one apsidal-line period for the three RSO test cases reported in Table 1. Note that the analytical computation only employs the initial osculating orbital elements as input and goes through the following steps:

1. The corresponding initial mean elements are computed with a Kozai-Lyddane transformation (Kozai (1959), Lyddane (1963)) accounting for first order J_2 short-periodic components and reported in Appendix B.

[5] The high-fidelity numerical propagator employed in this work is proprietary and not publicly released.

[6] The selection of order 23 was the result of extensive numerical testing. Such order balances accuracy with computational efficiency, and higher orders do not markedly change the results (Rivero et al., 2022).

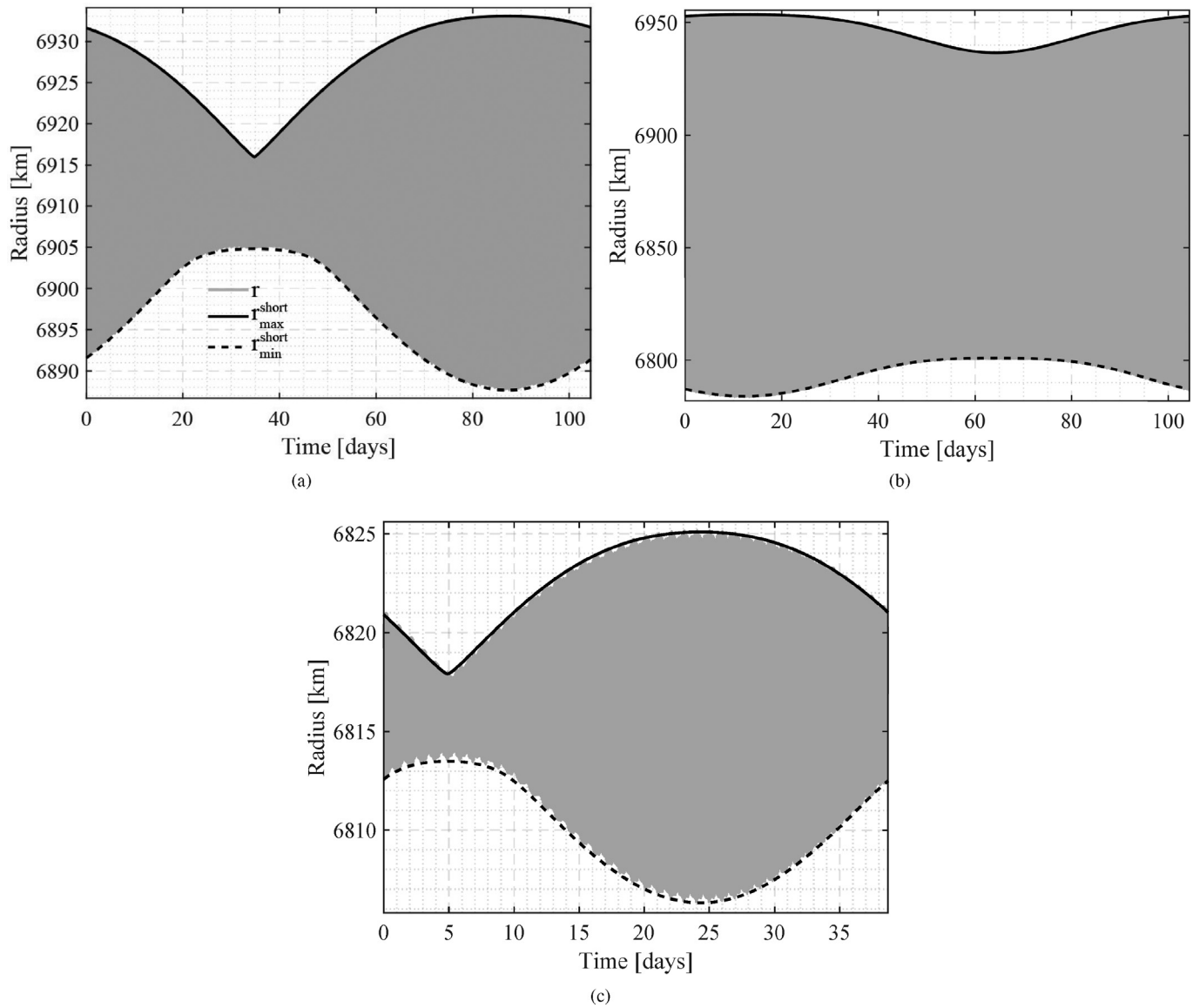

Fig. 4. Short-term space occupancy bounding for NORAD 47961 (a), NORAD 41460 (b) and NORAD 43518 (c).

2. The value of $\hat{e}(\tau)$, $\hat{\omega}(\tau)$, k are computed from Eqs. (4-11).
3. The maximum and minimum radius $\hat{r}$ corresponding to $\beta_1 = \beta(\tau)$ is computed, after solving the quartic equation of Eq. (34), from Eq. (35).

Fig. 4 suggests that the qualitative behavior of the long-period radial bounds fluctuations of different RSOs in LEO can be well reproduced with the proposed analytical model.

In order to fully assess the validity of the model an extensive test campaign has been conducted on a much wider number of test cases. The considered dataset included 16,972 orbits, filtered to exclude high eccentricity and high apogee orbits, and extracted from publicly available Two-line Elements (TLEs) data and using SGP4 theory (Vallado et al., 2006). These TLEs were only used to define the initial conditions of each orbit. Subsequently, each orbit was propagated using the high-fidelity numerical propagator for a time span equal to their precession period of the line of apsides, in order to generate the final dataset employed in this study.

To illustrate the distribution of eccentricity and semimajor axis for both the entire cataloged population and the final dataset considered in the study, Fig. 5 presents two histograms. The first histogram shows the distribution of eccentricities, and the second illustrates the distribution of semimajor axes. These histograms compare the overall cataloged objects (totaling 20,800 objects) with the subset selected after applying the filtering limitations. This approach provides a clearer understanding of the characteristics of the orbital parameters for the objects analyzed in this study.

For each orbit the minimum and maximum radii provided by the numerical propagation ($r^{num}_{max}, r^{num}_{min}$) across

Table 1
Initial osculating orbital elements of different test cases.

RSO	initial epoch [JD]	a [km]	e	i [deg]	ω [deg]
NORAD 47961	2,459,878.68	6,908.52	0.0025	97.50	65.67
NORAD 41460	2,459,880.68	6,875.35	0.0125	98.27	129.61
NORAD 43518	2,459,886.86	6,823.00	0.0006	34.93	289.07

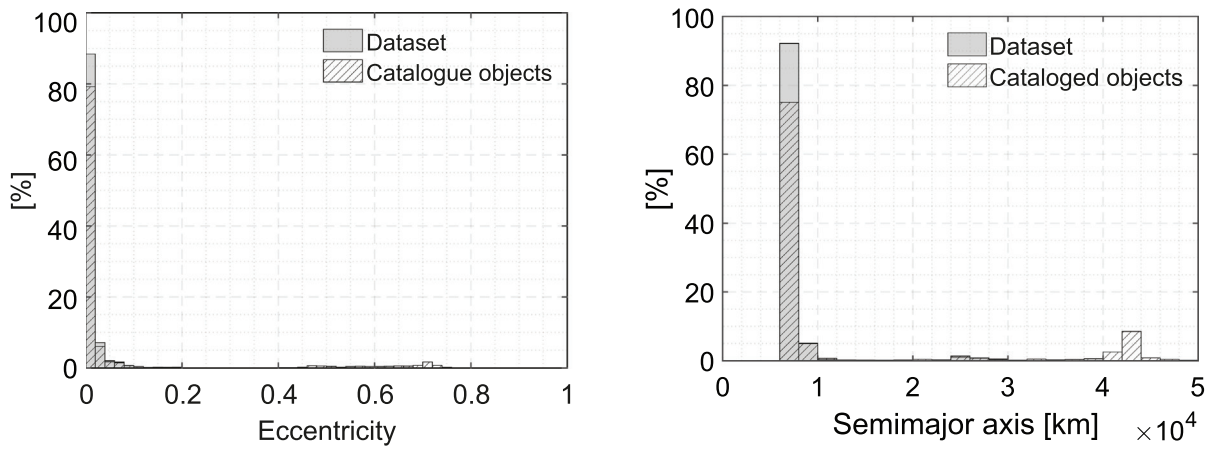

Fig. 5. Distribution of eccentricity (left) and semimajor axis (right) for the entire cataloged population and the filtered dataset considered in this study.

a 5-day window were compared to the corresponding values ($r_{max}^{short}, r_{min}^{short}$) of the analytical short-term SO model to compute the model error:

$$\varepsilon = \max\left\{\left|r_{max}^{short} - r_{max}^{num}\right|, \left|r_{min}^{short} - r_{min}^{num}\right|\right\}. \tag{37}$$

Fig. 6 displays the error distribution and its cumulative density function, revealing a mean error margin of approximately 0.5 km. Given the inherent uncertainties in TLE-derived data, this level of accuracy is deemed satisfactory.

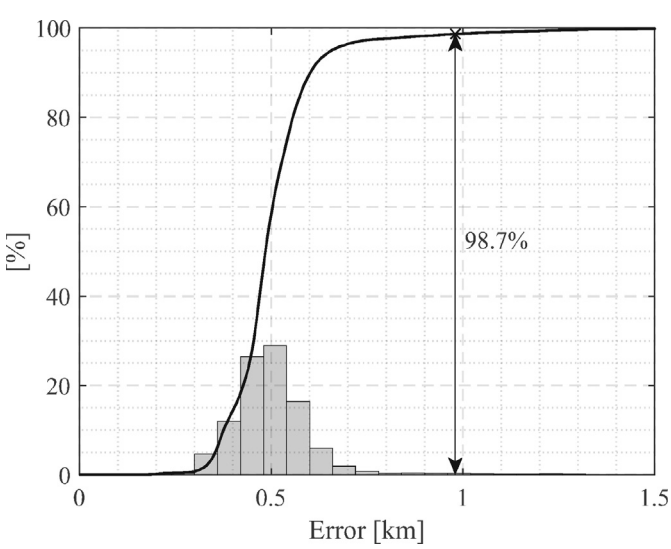

Fig. 6. Short-term SO error histogram and cumulative density function.

Remarkably, over 98.7% of the analyzed cases exhibited an error of less than one kilometer, highlighting the robustness and applicability of the model.

To further assess the model performance under limit orbital conditions, two limiting cases have been analyzed. For the orbit with the highest eccentricity in the dataset ($\hat{e} = 0.09949$), the error was 0.4320 km, which remains close to the average error observed across all cases. In contrast, the orbit with the largest semimajor axis ($\hat{a} = 31,153.96$ km) resulted in one of the highest errors, 1.1666 km, placing it within the less probable upper tail of the error distribution. These examples suggest that, although edge cases may present larger errors, the overall accuracy is influenced by a combination of orbital characteristics, not solely by eccentricity or semimajor axis.

3. Space occupancy conjunction filter

The relatively good accuracy of the proposed SO model motivates its application to first-stage conjunction filtering. More specifically, it is attractive to consider an SO-based filter as a replacement of the classical AP-filter, which typically serves as the initial filtering stage in several conjunction analysis tools (see for instance Alarcón Rodríguez et al. (2002)) and is based on the simple idea of eliminating pairs of objects with non-overlapping ranges of radii.

The idea is to employ the short-term SO model as a more accurate, yet computationally efficient tool in order to predict a radial overlap for a given conjunction screen-

ing horizon. In addition to the classical AP-filter (Hoots et al., 1984), three different implementations of a conjunction filter based on the SO model are considered:

1. "AP-filter". The classical filter where the radial bounds are defined by apogee and perigee radius.
2. "SO-filter". A filter implementation where the radial bounds are evaluated through Eq. (29), which involves the solution of the quartic equation, Eq. (34).
3. "SO-filter exact". A refined but computationally expensive procedure where the radial bounds are evaluated by solving Eq. (21).
4. "SO-filter raw". A very simple but crude implementation where the maximum and minimum radii are obtained by bounding the two brackets of Eq. (19) by their respective extrema (as described in Rivero et al. (2023)).

To test and compare the filters, the same dataset and high-fidelity model of Section 2.2.3 was employed. This set of orbits provides a total of 144 million pairs to analyze. The latest TLE available for each object between 10/24/2022 00:00:00 and 11/04/2022 11:30:00 was used to define the initial condition. All orbits were then propagated to a common epoch, t_0 (11/02/2022 09:18:20), corresponding to the mean epoch of the dataset. The mean propagation time to reach t_0 was approximately 17.5 h. The corresponding initial mean orbital elements have been derived using Kozai-Lyddane equations[7].

All computations were performed on a laptop equipped with an 11th Gen Intel(R) Core(TM) i7-11800H processor, with a base frequency of 2.30 GHz. The code was implemented and executed in MATLAB R2021b. Under these conditions, the computational times required by the filters to process 16,972 orbits were as follows: 7.20 min (AP-filter), 7.91 min (SO-filter raw), 8.30 min (SO-filter), and 12.74 min (SO-filter exact). Independently of the specific analysis, the inclusion or omission of the buffers leads to a negligible time difference, typically on the order of a few seconds. In relative terms, the SO-filter resulted in an increase in computation time of only 14.90% compared to the AP-filter, while the SO-filter raw was only slightly more demanding, with a 9.80% increase. On the other hand, the exact SO-filter showed a more substantial increase of 76.86%.

Next, the different filters were compared in terms of accuracy and effectiveness—understood as the balance between false positives and false negatives—without applying any correction buffer in Section 3.1. Correction buffers (see Alfano and Finkleman (2014)) were then applied in order to fully remove false negative errors and Section 3.2 provides a comparative analysis of the performance of the modified filters. Finally, Section 3.3 describes the analysis of the effect of atmospheric drag.

3.1. Comparison of filters without buffer

A simple and preliminary assessment of the performance of the filters has been conducted considering the same 16,972 RSOs dataset as in Section 2.2.3 over a 5-day time horizon and without applying any correction buffer. The high-fidelity propagator described in that section is employed to obtain the numerically propagated ("true") results in terms of radial bounds to be compared to the ones employed by the different filters. Note that, for simplicity, the effect of atmospheric drag has been neglected in this analysis and will be dealt with in a dedicated section (Section 3.3).

Following a widely adopted terminology, a *positive* outcome refers to a pair of orbits having overlapping ranges of radii, which is a necessary but not sufficient condition for a collision to occur. Conversely, a *negative* outcome corresponds to non-overlapping ranges of radii, which (in a deterministic analysis) rules out the possibility of a collision within the established time window.

Given two orbits i and j, each filter excludes the pair (i, j) when at least one of the following two conditions is met:

$$\begin{aligned} r_{max,i} &< r_{min,j}, \\ r_{max,j} &< r_{min,i}, \end{aligned} \tag{38}$$

where r_{max} and r_{min} are the radial bounds computed by a given filter and solely based on the state vectors of the two objects at the reference time t_0. Specifically, for the AP-filter the mean apogee and perigee at t_0 were employed since using osculating values was seen to deteriorate the filter performance. This can be appreciated in Fig. 3: the osculating apogee and perigee are highly time-dependent.

For a given filter, a *false positive* error occurs whenever the filter predicts a radial range overlap that does not take place in that time window while a *false negative* error occurs whenever the filter fails to predict a radial range overlap and filters out an object pair that could collide. A visual comparison of real vs. filter-predicted positive outcomes is represented in Fig. 7.

From the total number of pairs (N), the number of pairs eliminated by the filter (N_{out}), the number of real positive outcomes obtained numerically[8] (N_{RP}) and the number of false positives (N_{FP}) and false negatives (N_{FN}) scored by the filter, one can define the *false positives to detected real positives* ratio as:

$$\rho_{FP} = \frac{N_{FP}}{N_{RP} - N_{FN}}, \tag{39}$$

[7] Note that implementing the filter based on the average orbital elements extracted directly from the TLEs provided considerably worse performance.

[8] Real positives are determined by computing the maximum and minimum radius values using the high-fidelity propagation. Pairs that exhibit overlap within the altitude band based on these values are classified as real positives.

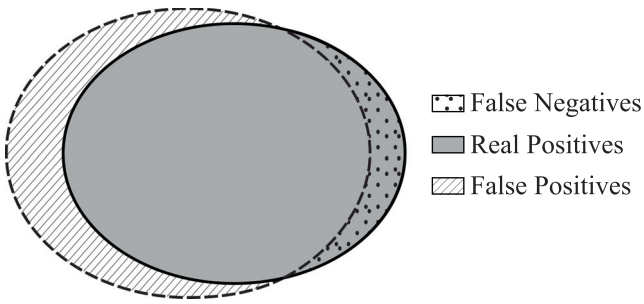

Fig. 7. Schematic of real (solid line) and filter-predicted (dash-line) positive outcomes.

the *false negatives to detected real positives* ratio as:

$$\rho_{FN} = \frac{N_{FN}}{N_{RP} - N_{FN}}, \tag{40}$$

and the *filter effectiveness* as:

$$\eta = \frac{N_{out}}{N}. \tag{41}$$

These coefficients, ρ_{FP} and ρ_{FN}, range from 0 to infinity. A value of 0 corresponds to the best-case scenario, where there are no false positives or false negatives, while a value approaching infinity represents the worst-case scenario, where the filter fails to detect any true positives and incorrectly flags all true negatives as positives.

Table 2 summarizes the results of the four filters using the three metrics previously defined.

First of all, it should be noted that the less computationally expensive SO-filter almost matches the performance of the SO-filter exact in spite of the simplifying assumptions. In relation to the false positives, the conservative character of the SO-filter raw makes it perform even poorer than the AP-filter whereas the SO-filter manages to reduce them by two orders of magnitude. On the other hand, compared to the AP-filter, both the SO-filter and SO-filter raw considerably reduce the number of false negatives.

3.2. Comparison of filters with buffer

In conjunction analysis, false negative errors must be avoided since they can leave potentially dangerous conjunctions undetected. To eliminate false negatives, buffer distances are employed to ensure that a given filter correctly bounds the radius evolution of each object in the catalogue even at the expense of increasing false positives.

After adding buffer distances b_i and b_j, a given filter excludes from further conjunction assessment all pairs of satellites obeying any of the two conditions:

$$\begin{aligned} r_{max,i} + b_i &< r_{min,j} - b_j, \\ r_{max,j} + b_j &< r_{min,i} - b_i. \end{aligned} \tag{42}$$

Ideally, for a given filter, the applied buffer size should be as small as possible and with minimum sensitivity to the specific initial conditions of the object considered.

In order to establish the appropriate size of a buffer for a given filter, extensive testing needs to be conducted using the full catalogue of objects and a high fidelity model as the one described in Section 2.2.3. As seen in Fig. 4, the errors in computing the radius bounds of an individual RSO vary considerably (depending on the initial epoch) throughout one full line of apsides precession period. This means that a conservative value needs to be computed taking the maximum buffer size over that period[9].

Fig. 8 presents the statistics of the bounding error reduction as a function of the applied buffer for each of the four filters considered and using the same 16,972 RSOs population. In this context, the bounding error, denoted as ε^*, is defined as the discrepancy between the estimated and actual bounds when the estimation underestimates the true range. Specifically, it is computed as:

$$\varepsilon^* = \max\left\{r_{max}^{num} - r_{max}, r_{min} - r_{min}^{num}\right\}. \tag{43}$$

The results displayed in the left-hand side of the figure take into account a 5-day time span while the ones in the right-hand side are based on a full line of apsides precession period (a few months). The reference initial epoch is $JD = 2,459,885.89$ for both cases. It can be seen that the behavior is similar; however, extending the propagation over a precession period moves the histogram to the right thereby increasing the buffers. The plots show a typical 9 km required buffer for the AP-filter (3–6 km for the 5-day time span) compared to a 0.5 km for the SO-filters (0.1–0.2 km for the 5-day time span). A complete elimination of bounding error for the AP-filter requires a buffer of 11.527 km (10.722 km for the 5-day time span) while the SO-filter requires a buffer of 2.508 km (1.119 km for the 5-day time span).

The preceding results suggest a possible improvement in terms of buffer minimization for the SO-filter. The fact that the maximum required buffer is five times greater than the statistical mode of its distribution leads to the consideration of whether the few cases requiring a larger buffer share some common characteristics and may be treated differently.

Indeed, it has been observed that these few cases are characterized by more eccentric and higher altitude orbits. Therefore, in order to avoid penalizing the entire population with such a high buffer, it has been decided to divide the population into six distinct groups based on the minimum altitude and eccentricity and apply a bespoke buffer to each category for each filter. Regarding eccentricity, the population has been divided into two groups. Within the low eccentricity orbits, four categories have been distinguished based on minimum initial mean altitude: the first one with a minimum altitude below 400 km, the second group with a minimum altitude between 400 and 700 km, the third group between 700 and 1,000 km, and finally,

[9] For those orbits for which this time interval exceeded one year the evolution has been calculated over 365 days.

Table 2
Summary of filter performance comparison, without buffer.

Filter	Real Positives	False Positives	False Negatives	ρ_{FP}	ρ_{FN}	η
AP-filter	31,950,589	224,316	537,463	0.714 %	1.711 %	77.659 %
SO-filter	31,950,589	2,242	64,006	0.007 %	0.201 %	77.813 %
SO-filter exact	31,950,589	1,889	67,054	0.006 %	0.210 %	77.813 %
SO-filter raw	31,950,589	445,342	6,752	1.394 %	0.021 %	77.505 %

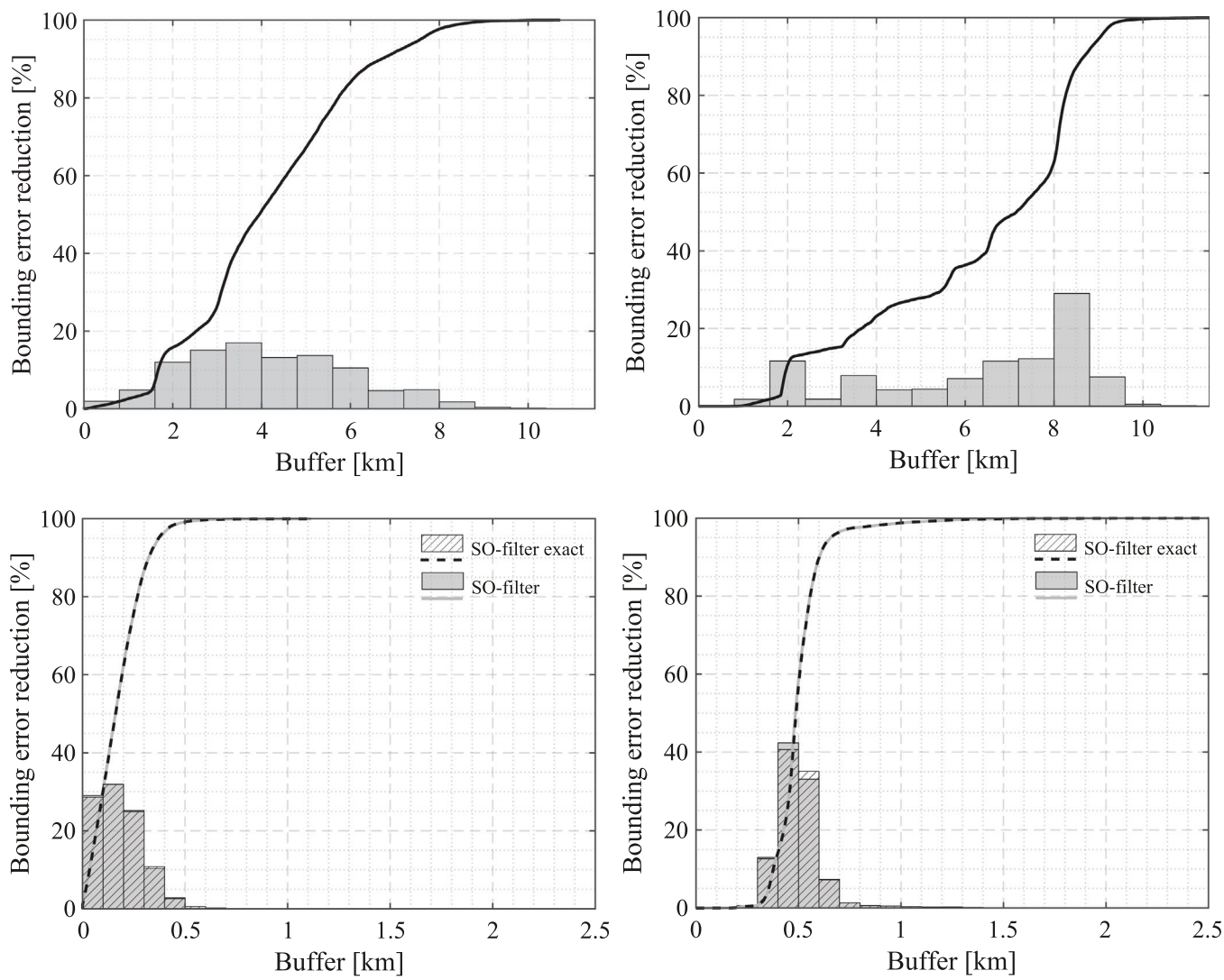

Fig. 8. Statistics of the bounding error reduction as a function of the applied buffer for the AP-filter (top) and the SO-filter (bottom) and for a 5-day (left) and full precession period time span (right).

higher altitude orbits, including all those above 1,000 km. On the other hand, orbits with higher eccentricities have been divided into two groups: low-altitude orbits with a minimum altitude below 1,000 km, and high-altitude orbits comprising all others. Table 3 presents the required buffer for each filter and each category.

Table 4 summarizes the results of the four filters with their buffers presented in Table 3. It is important to highlight that this latest improvement represents a decrease of at least 50% in false positives for all the SO-filters, removing almost 800,000 more pairs at a very small additional cost when compared to the case of a common applied buf-

Table 3
Required buffer for each filter according to categories.

Eccentricity	Altitude [km]	SO-filter [km]	AP-filter [km]	SO-filter exact [km]	SO-filter raw [km]
$e \leqslant 0.01$	$r_{min} - R_{\oplus} < 400$	0.9782	11.5271	0.9780	1.2586
	$r_{min} - R_{\oplus} < 700$	1.2823	11.2849	1.2827	3.3474
	$r_{min} - R_{\oplus} < 1,000$	0.7066	10.2531	0.7066	3.0379
	$r_{min} - R_{\oplus} > 1,000$	2.0260	8.5749	2.0263	2.8355
$0.01 < e \leqslant 0.1$	$r_{min} - R_{\oplus} < 1,000$	0.9009	10.7209	0.9018	3.0047
	$r_{min} - R_{\oplus} > 1,000$	2.5072	8.4504	2.5076	2.7253

Table 4
Summary of filter performance comparison with the required buffer.

Filter	Real Positives	False Positives	False Negatives	ρ_{FP}	η
AP-filter	31,950,589	5,508,159	0	17.240 %	73.990 %
SO-filter	31,950,589	527,018	0	1.649 %	77.449 %
SO-filter exact	31,950,589	524,183	0	1.641 %	77.451 %
SO-filter raw	31,950,589	2,145,317	0	6.714 %	76.325 %

fer. Moreover, it is worth noting that all configurations achieve zero false negatives, albeit at the expense of a considerable increase in false positives compared to the no-buffer case.

To close the comparison, the behavior of the ρ_{FP} and ρ_{FN} ratios with respect to the buffer size applied to the SO-filter and the AP-filter has been analyzed in a similar way as done in Alfano and Finkleman (2014).

Fig. 9 illustrates the relationship between the ρ_{FP} and buffer size. As expected, the false positive ratio increases with larger buffer sizes. Interestingly, the SO-filter outperforms the AP-filter for buffer sizes below 1 km, but beyond this threshold, the AP-filter demonstrates superior performance.

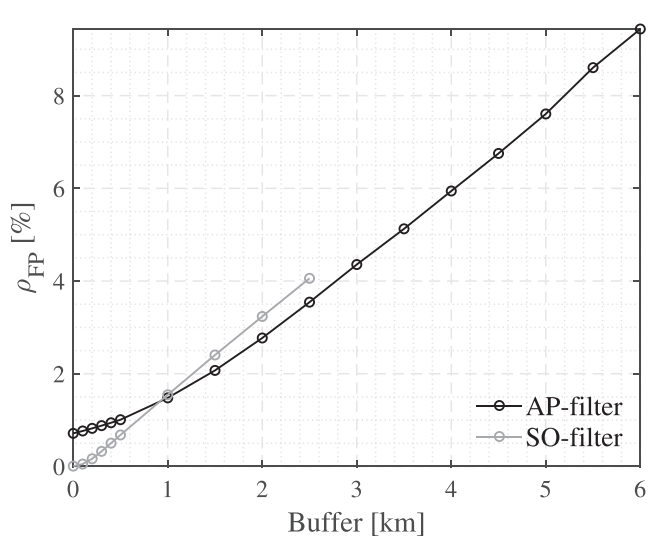

Fig. 9. False positive to detected real positive ratio as a function of the applied buffer.

Fig. 10 plots the ρ_{FN} against the buffer. This graph demonstrates the superior performance of the SO filter. Notably, a buffer of 0.5 km is sufficient to eliminate all false negatives when using the SO-filter, whereas the AP-filter necessitates a buffer exceeding 6 km to achieve the same outcome.

Finally, Fig. 11 plots the fraction of detected real positives:

$$\frac{1}{\rho_{RP}} = \frac{N_{RP} - N_{FN}}{N_{RP}} = \frac{1}{1 + \rho_{FN}}, \tag{44}$$

against the previously defined false positive to detected real positive rate ρ_{FP} (Eq. (39)) showing what is known as *receiver operating characteristic curve* (Fawcett, 2006). This type of curve is used to compare two operating characteristics and depicts relative trade-offs between benefits (true positives) and costs (false positives). The filter performance, which is higher the closer this curve is to the upper left corner, is clearly much higher for the SO-filter case than the classical AP-filter.

3.3. Atmospheric drag

The previous results do not take into account the effect of atmospheric drag, which is relevant only for relatively low altitude orbits (i.e., below 500 km). In order to extend the SO analysis and filter design to include this effect what can be done is to add a lower altitude buffer to account for the drag-induced decay with a simplified model that allows a quick estimation to be included in the filter estimates. A simple recipe to do that is to exploit the starred ballistic coefficient (B^*) publicly available from TLEs data. The following analysis investigates this possibility to extend the SO model accordingly, and computes the updated filter performance metrics.

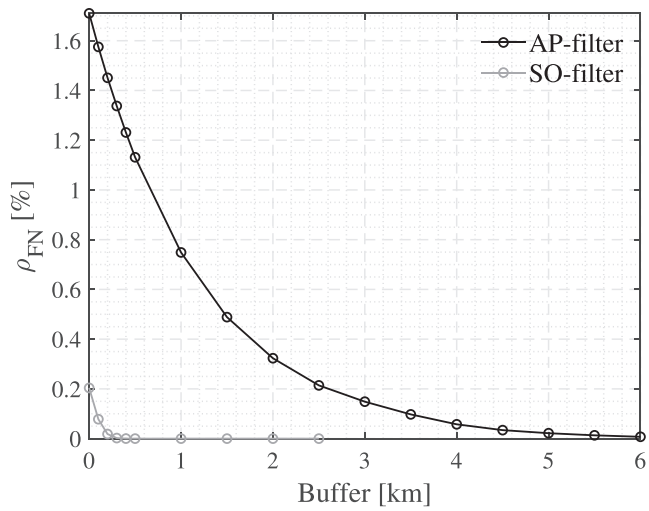

Fig. 10. False positive to detected real positive ratio as a function of the applied buffer.

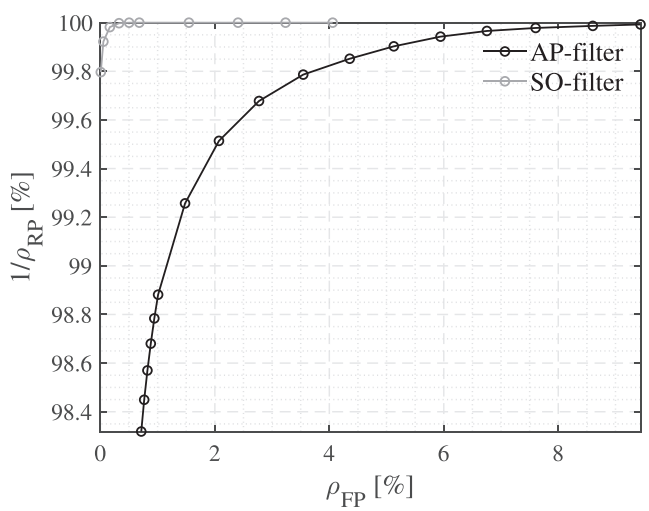

Fig. 11. Operating characteristic curve.

The decrease in minimum altitude as a function of time for a near circular orbit can be estimated from Eq. (45) developed by Billik (Karrenberg, 1964):

$$h = \frac{1}{\beta_a} \log \left[e^{\beta_a h_0} - B\sqrt{\mu R_\oplus}\beta_a \bar{\rho} t \right], \tag{45}$$

where B is the (inverse) ballistic coefficient, h_0 is the initial altitude, t is time, and $\beta_a, \bar{\rho}$ are coefficients characterizing an exponential density model according to:

$$\rho = \bar{\rho} e^{-\beta_a h}. \tag{46}$$

The classical definition for the (inverse) ballistic coefficient is given in terms of the drag coefficient of the satellite C_D, the frontal area A_v and the satellite mass, m_v as:

$$B = \frac{C_D A_v}{m_v}. \tag{47}$$

From Vallado (2001), the classical ballistic coefficient can be related to the starred ballistic coefficient readily available from TLEs as:

$$B = 12.741621 B^*. \tag{48}$$

The computation of the two coefficients of the exponential density model was carried out using the procedure explained in Karrenberg (1964). This involves substituting two pairs of data, altitude, and density, into Eq. (46) and solving the system to obtain the values of the coefficients. A layered model approach was employed, computing β_a and $\bar{\rho}$ for each 50 km altitude layer. Altitude and density data were obtained from the exponential model in Vallado (2001), utilizing the data from Table 8–4. Table 5 displays the coefficient values for each layer.

Before incorporating Eq. (45) into the filter a numerical test campaign has been conducted to verify that it is sufficiently conservative. The analytically computed altitude drop has been compared with the results of a numerical integration based on the high-fidelity model of Section 2.2.3 now including the atmospheric drag (using the exponential density model presented in Vallado (2001), for consistency). The same dataset of 16,972 RSOs orbits considered in Section 2.2.3 has been numerically propagated over a 5-day timespan and considering the B* drag coefficient (Eq. (48)) as provided in the TLEs catalog. The decrease in alti-

Table 5
Coefficients characterizing the exponential density model (Vallado, 2001).

Minimum altitude [km]	β_a [1/km]	$\bar{\rho}$ [km/m^3]
$r_{min} - R_\oplus < 175$	0.0549	8.059×10^{-06}
$r_{min} - R_\oplus < 225$	0.0404	6.426×10^{-07}
$r_{min} - R_\oplus < 275$	0.0220	1.013×10^{-08}
$r_{min} - R_\oplus < 325$	0.0186	4.078×10^{-09}
$r_{min} - R_\oplus < 375$	0.0195	5.440×10^{-09}
$r_{min} - R_\oplus < 425$	0.0163	1.629×10^{-09}
$r_{min} - R_\oplus < 500$	0.0164	1.716×10^{-09}

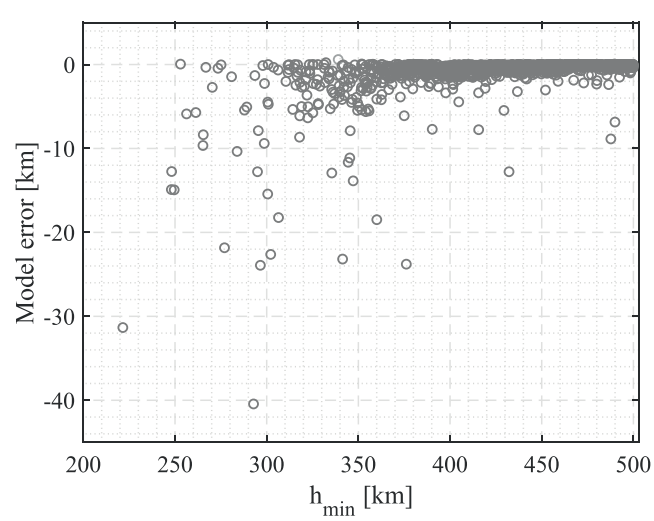

Fig. 12. Model error with respect to the minimum real altitude.

Table 6
Summary of filter performance comparison including atmospheric drag.

Filter	Real Positives	False Positives	False Negatives	ρ_{FP}	ρ_{FN}	η
AP-filter	31,950,589	5,517,972	0	17.270 %	0 %	73.983 %
SO-filter	31,950,589	537,612	0	1.683 %	0 %	77.441 %
SO-filter exact	31,950,589	534,708	0	1.674 %	0 %	77.443 %
SO-filter raw	31,950,589	2,155,811	0	6.747 %	0 %	76.318 %

tude predicted by the analytical model (i.e., the value of h computed using Eq. (45)) is compared with the result obtained through high-fidelity numerical propagation, h_{num}. Fig. 12 summarizes the difference between the two (excluding reentered objects, i.e. with altitudes reaching below 150 km), which effectively represents the model error defined as $h - h_{num}$. It can be seen that the model overestimates the descent in the great majority of cases. Nevertheless, to make the filter fully conservative, a safety margin of 0.6 km has been added to ensure that the bounds used cover the entire space occupancy of the orbits, avoiding false negatives. In those cases where the model predicts reentry, the minimum radius has been set to 0, since the study of reentry is out of the scope of this analysis.

In order to analyze and compare the performance of the SO-filter with the added atmospheric drag correction, the results of Table 4 have been recomputed. Note that a drag correction is only considered for RSOs below 500 km altitude. Table 6 summarizes the new results.

It can be seen that the ρ_{FP} increases slightly compared to the previous analysis, by only 0.03–0.04%. Additionally, it is evident that the atmospheric drag affects the same way the four filters confirming the conclusions of Section 3.2.

4. Conclusions

This work introduces the concept of short-term space occupancy, a generalization of the space occupancy framework that improves accuracy for applications with short time horizons, such as space situational awareness and conjunction assessment. The short-term SO model is based on a zonal problem formulation and enables the precise estimation of an object-occupied altitude range over a given time interval by solving a quartic equation.

Building upon this theoretical foundation, the SO-filter is proposed. It is a novel conjunction filter that leverages the short-term SO model to more accurately identify object pairs with overlapping radial ranges while accounting for the effects of zonal harmonics. The performance of the SO-filter is compared to the classical AP-filter and two alternative SO-filter implementations using a large dataset of space objects and a high-fidelity propagator.

The results demonstrate that the SO-filter substantially outperforms the AP-filter in terms of both false positive and false negative rates. Without applying correction buffers, the SO-filter reduces false positives by two orders of magnitude and yields eight times fewer false negatives compared to the AP-filter. When correction buffers are introduced to eliminate false negatives, the SO-filter requires a buffer three times smaller than the AP-filter, resulting in a tenfold reduction in false positives. The performance of the SO-filter is comparable to the more computationally expensive exact SO-filter implementation, making it an attractive option for efficient and accurate conjunction screening.

Future work will focus on extending the applicability of the current methodology to orbits with higher eccentricity, particularly through the exploration of a third-order radius expansion. This will be investigated as a potential tool to extend the filter's performance to every objects in LEO region. Additionally, this effort will be complemented by the expansion of these concepts to the classical orbit path and time filters by continuing to exploit the solutions of the zonal-perturbed two-body problem. In addition, future extensions could explore the integration of these methods into existing space traffic management systems and investigate their potential applications in other areas, such as mission planning and space debris remediation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

R. Vazquez and A. S. Rivero gratefully acknowledge support by grant PID2023-147623OB-I00 funded by MICIU/AEI/10.13039/ 501100011033 and by "ERDF A way of making Europe". C. Bombardelli gratefully acknowledges support provided by MINECO/AEI and FEDER/EU under Project PID2020-112576 GB-C21. A. S. Rivero was funded by a FPU grant from the Spanish Ministry of Universities.

The authors thank Prof. Giulio Baù from the University of Pisa for revising and correcting the derivations of Appendix A.

The authors acknowledge the use of artificial intelligence tools to assist with English language editing, proper formatting, and grammar correction.

Appendix A. Global bounds of the short-term SO

This appendix provides a detailed examination of the mathematical derivation required to identify critical points, essential for determining the extremal values of the function r within the short-term SO problem, $\left(\hat{\theta}_i^*, \hat{\beta}_i^*\right)$. It also includes an exhaustive analysis to determine the type of each critical point (maximum, minimum, or saddle point) required to ascertain the global maximum and minimum $\left(\hat{\theta}_{max}^*, \hat{\beta}_{max}^*\right)$ and $\left(\hat{\theta}_{min}^*, \hat{\beta}_{min}^*\right)$, as well as local extrema, $\left(\hat{\theta}_{Lmax}^*, \hat{\beta}_{Lmax}^*\right)$ and $\left(\hat{\theta}_{Lmin}^*, \hat{\beta}_{Lmin}^*\right)$.

The analysis primarily centers on computing the first and second derivatives with respect to both variables, elucidating the fundamental principles underpinning the computation of critical points. Specifically, the first derivatives are detailed in Eqs. (49) and (50), while the second derivatives are presented in Eqs. (51)–(53).

$$\frac{\partial \hat{r}}{\partial \beta} = -\hat{a}e_p \sin\left(\hat{\theta} - \beta\right), \tag{49}$$

$$\frac{\partial \hat{r}}{\partial \hat{\theta}} = \hat{a}\left(e_p \sin\left(\hat{\theta} - \beta\right) - e_f \cos\hat{\theta}\right) - \frac{J_2 \sin^2 \hat{i}}{2\hat{a}} \sin\left(2\hat{\theta}\right), \tag{50}$$

$$\frac{\partial^2 \hat{r}}{\partial \beta^2} = \hat{a}e_p \cos\left(\hat{\theta} - \beta\right), \tag{51}$$

$$\frac{\partial^2 \hat{r}}{\partial \beta \partial \hat{\theta}} = -\hat{a}e_p \cos\left(\hat{\theta} - \beta\right), \tag{52}$$

$$\frac{\partial^2 \hat{r}}{\partial \hat{\theta}^2} = \hat{a}\left(e_p \cos\left(\hat{\theta} - \beta\right) + e_f \sin\hat{\theta}\right) - \frac{J_2 \sin^2 \hat{i}}{\hat{a}} \cos\left(2\hat{\theta}\right). \tag{53}$$

Thus, to find $\hat{\theta}_i^*$ and β_i^*, which represent the candidates for the extrema, one needs to solve:

$$0 = -\hat{a}e_p \sin\left(\hat{\theta}_i^* - \beta_i^*\right), \tag{54}$$

$$0 = \hat{a}\left(e_p \sin\left(\hat{\theta}_i^* - \beta_i^*\right) - e_f \cos\hat{\theta}_i^*\right) - \frac{J_2 \sin^2 \hat{i}}{2\hat{a}} \sin\left(2\hat{\theta}_i^*\right). \tag{55}$$

Initially, Eq. (54) yields two possibilities: $\hat{\theta}_i^* = \beta_i^*$ or $\hat{\theta}_i^* = \pi + \beta_i^*$. Substituting these values into Eq. (55) results in the following equation:

$$\begin{aligned} 0 &= -\hat{a}e_f \cos\hat{\theta}_i^* - \frac{J_2 \sin^2 \hat{i}}{2\hat{a}} \sin\left(2\hat{\theta}_i^*\right) \\ &= -\cos\left(\hat{\theta}_i^*\right)\left[\hat{a}e_f + \frac{J_2 \sin^2 \hat{i}}{\hat{a}} \sin\left(\hat{\theta}_i^*\right)\right], \end{aligned} \tag{56}$$

where the expansion of the sine of a double angle has been taken into account. The possibility $\cos\left(\hat{\theta}_i^*\right) = 0$ is then considered, leading to $\hat{\theta}_i^* = \pi/2$ or $\hat{\theta}_i^* = -\pi/2$. This consideration results in four critical points:

$$\left(\hat{\theta}_i^*, \beta_i^*\right) = \{(\pi/2, \pi/2), (-\pi/2, \pi/2), (-\pi/2, -\pi/2), (\pi/2, -\pi/2)\}. \tag{57}$$

Additionally, there exist other critical points determined by the values of $\hat{\theta}_i^*$ such that $\hat{a}e_f + \frac{J_2 \sin^2 \hat{i}}{\hat{a}} \sin\left(\hat{\theta}_i^*\right) = 0$. When $\left|\frac{\hat{a}^2 e_f}{J_2 k}\right| > 1$, there are no solutions of this kind; however, whether $\left|\frac{\hat{a}^2 e_f}{J_2 k}\right| < 1$, there exist four additional critical points:

$$\begin{aligned} \left(\hat{\theta}_i^*, \beta_i^*\right) \quad = \Big\{ &\left(-\arcsin\left(\frac{\hat{a}^2 e_f}{J_2 \sin^2 \hat{i}}\right), -\arcsin\left(\frac{\hat{a}^2 e_f}{J_2 \sin^2 \hat{i}}\right)\right), \left(\pi - \arcsin\left(\frac{\hat{a}^2 e_f}{J_2 \sin^2 \hat{i}}\right), -\arcsin\left(\frac{\hat{a}^2 e_f}{J_2 \sin^2 \hat{i}}\right)\right), \\ &\left(-\pi + \arcsin\left(\frac{\hat{a}^2 e_f}{J_2 \sin^2 \hat{i}}\right), -\pi + \arcsin\left(\frac{\hat{a}^2 e_f}{J_2 \sin^2 \hat{i}}\right)\right), \left(\arcsin\left(\frac{\hat{a}^2 e_f}{J_2 \sin^2 \hat{i}}\right), -\pi + \arcsin\left(\frac{\hat{a}^2 e_f}{J_2 \sin^2 \hat{i}}\right)\right)\Big\}. \end{aligned} \tag{58}$$

Computing now the Hessian H,

$$H\left(\hat{\theta}, \beta\right) = \begin{bmatrix} \hat{a}e_p \cos\left(\hat{\theta} - \beta\right) & -\hat{a}e_p \cos\left(\hat{\theta} - \beta\right) \\ -\hat{a}e_p \cos\left(\hat{\theta} - \beta\right) & \hat{a}\left(e_p \cos\left(\hat{\theta} - \beta\right) + e_f \sin\hat{\theta}\right) - \frac{J_2 k}{\hat{a}} \cos\left(2\hat{\theta}\right) \end{bmatrix}. \tag{59}$$

One has to remember that a critical point would be a saddle point if there is no sign definiteness. To have sign definiteness, it is required that the determinant of the Hessian is positive. Now, the determinant can be obtained as

$$\det(H) = \hat{a}e_p \cos\left(\hat{\theta} - \beta\right)\left(\hat{a}e_f \sin\hat{\theta} - \frac{J_2 \sin^2 \hat{i}}{\hat{a}} \cos\left(2\hat{\theta}\right)\right). \tag{60}$$

At the critical points, the situation unfolds as follows:

$$H(\pi/2,\pi/2)=\begin{bmatrix}\hat{a}e_p & -\hat{a}e_p\\ -\hat{a}e_p & \hat{a}\left(e_p+e_f\right)+\frac{J_2\sin^2\hat{i}}{\hat{a}}\end{bmatrix},\quad \det(H)=\hat{a}^2e_pe_f+e_pJ_2\sin^2\hat{i}, \tag{61}$$

$$H(\pi/2,-\pi/2)=\begin{bmatrix}-\hat{a}e_p & \hat{a}e_p\\ \hat{a}e_p & \hat{a}\left(-e_p+e_f\right)+\frac{J_2\sin^2\hat{i}}{\hat{a}}\end{bmatrix},\quad \det(H)=-\hat{a}^2e_pe_f-e_pJ_2\sin^2\hat{i}, \tag{62}$$

$$H(-\pi/2,-\pi/2)=\begin{bmatrix}\hat{a}e_p & -\hat{a}e_p\\ -\hat{a}e_p & \hat{a}\left(e_p-e_f\right)+\frac{J_2\sin^2\hat{i}}{\hat{a}}\end{bmatrix},\quad \det(H)=-\hat{a}^2e_pe_f+e_pJ_2\sin^2\hat{i}, \tag{63}$$

$$H(-\pi/2,\pi/2)=\begin{bmatrix}-\hat{a}e_p & \hat{a}e_p\\ \hat{a}e_p & \hat{a}\left(-e_p-e_f\right)+\frac{J_2\sin^2\hat{i}}{\hat{a}}\end{bmatrix},\quad \det(H)=\hat{a}^2e_pe_f-e_pJ_2\sin^2\hat{i}, \tag{64}$$

$$H\left(\hat{\theta}_5^*,\beta_5^*\right)=\begin{bmatrix}\hat{a}e_p & -\hat{a}e_p\\ -\hat{a}e_p & \hat{a}e_p-\frac{J_2^2\sin^4\hat{i}-\hat{a}^4e_f^2}{\hat{a}J_2\sin^2\hat{i}}\end{bmatrix},\quad \det(H)=\frac{e_p}{J_2\sin^2\hat{i}}\left(\hat{a}^4e_f^2-J_2^2\sin^4\hat{i}\right), \tag{65}$$

$$H\left(\hat{\theta}_6^*,\beta_6^*\right)=\begin{bmatrix}-\hat{a}e_p & \hat{a}e_p\\ \hat{a}e_p & -\hat{a}e_p-\frac{J_2^2\sin^4\hat{i}-\hat{a}^4e_f^2}{\hat{a}J_2\sin^2\hat{i}}\end{bmatrix},\quad \det(H)=-\frac{e_p}{J_2\sin^2\hat{i}}\left(\hat{a}^4e_f^2-J_2^2\sin^4\hat{i}\right), \tag{66}$$

$$H\left(\hat{\theta}_7^*,\beta_7^*\right)=\begin{bmatrix}\hat{a}e_p & -\hat{a}e_p\\ -\hat{a}e_p & \hat{a}e_p-\frac{J_2^2\sin^4\hat{i}-\hat{a}^4e_f^2}{\hat{a}J_2\sin^2\hat{i}}\end{bmatrix},\quad \det(H)=\frac{e_p}{J_2\sin^2\hat{i}}\left(\hat{a}^4e_f^2-J_2^2\sin^4\hat{i}\right), \tag{67}$$

$$H\left(\hat{\theta}_8^*,\beta_8^*\right)=\begin{bmatrix}-\hat{a}e_p & \hat{a}e_p\\ \hat{a}e_p & -\hat{a}e_p-\frac{J_2^2\sin^4\hat{i}-\hat{a}^4e_f^2}{\hat{a}J_2\sin^2\hat{i}}\end{bmatrix},\quad \det(H)=-\frac{e_p}{J_2\sin^2\hat{i}}\left(\hat{a}^4e_f^2-J_2^2\sin^4\hat{i}\right). \tag{68}$$

An analysis of the type of critical points will be conducted, distinguishing between maxima, minima, and saddle points. In cases where $\left|\frac{\hat{a}^2e_f}{J_2k}\right|>1$, only the initial four critical points are relevant, with implications dependent on the sign of e_f. Typically positive for geocentric orbits, the value of e_f may turn negative very near to critical inclinations. Therefore, both scenarios should be analyzed.

When the frozen eccentricity is positive, the points $\left(\hat{\theta}_i^*,\beta_i^*\right)=(\pi/2,\pi/2),(-\pi/2,\pi/2)$ are the only ones that do not lead to a saddle point. Additionally, analysis of the sign of the element (1,1) of the Hessian matrix indicates that the maximum occurs at $\hat{\theta}_{max}^*=-\pi/2$, and the minimum at $\hat{\theta}_{min}^*=\pi/2$.

Conversely, in the event of a negative frozen eccentricity, the determinant of the Hessian matrix reverses polarity. Consequently, in this scenario, $\left(\hat{\theta}_i^*,\beta_i^*\right)=(\pi/2,-\pi/2)$, $(-\pi/2,-\pi/2)$ are the points where the extrema are reached. Considering the sign of the element (1,1) of the Hessian matrix, one can deduce that the maximum occurs at $\hat{\theta}_{max}^*=\pi/2$, while the minimum is obtained at $\hat{\theta}_{min}^*=-\pi/2$.

When $\left|\frac{\hat{a}^2e_f}{J_2k}\right|<1$, all eight positions become critical points, independent of the sign of e_f. Among the first four candidates, $\left(\hat{\theta}_{Lmin}^*,\beta_{Lmin}^*\right)=(\pi/2,\pi/2),(-\pi/2,-\pi/2)$ are the only points that do not lead to a saddle point. Based on the sign of the element (1,1) of the Hessian matrix, the conclusion is that both of them are minima. Considering the other four critical points, $\left(\hat{\theta}_{Lmax}^*,\beta_{Lmax}^*\right)=\left(\hat{\theta}_6^*,\beta_6^*\right),\left(\hat{\theta}_8^*,\beta_8^*\right)$ are the only ones that do not lead to a saddle point. Further examination of the sign of the element (1,1) leads to the conclusion that both represent maxima.

Note that in cases where only the first four positions are critical points, the function has only one maximum and one minimum, which corresponds to the globals, representing the solution of the long-term SO problem (evaluating the function r at these points yields the expressions of Eq. (16) and (17)). Conversely, in instances where all eight positions become critical points, there are two maxima and two minima. Evaluating r at each position yields the global maximum, $\left(\hat{\theta}_{max}^*,\beta_{max}^*\right)$, and minimum, $\left(\hat{\theta}_{min}^*,\beta_{min}^*\right)$, representing the solution of the long-term SO[10]. However, the other extrema, which are local, $\left(\hat{\theta}_{Lmax}^*,\beta_{Lmax}^*\right)$, $\left(\hat{\theta}_{Lmin}^*,\beta_{Lmin}^*\right)$, must also be considered for the analysis of the short-term SO problem.

Appendix B. Osculating to mean orbital elements conversions

The algorithm employed to convert the osculating orbit elements into mean orbit elements is outlined in this appendix. It is based on the theory developed by Kozai (1959) and Lyddane (1963).

This mapping translates any osculating (instantaneous) orbital elements into mean (orbit averaged, with short period motion removed) orbital element equivalent values. It is important to take into account that only first order J_2

[10] Note that this maximum solution does not correspond to the one obtained in Bombardelli et al. (2021), as this work does not account for these particular cases. These cases stem from the inclusion of additional harmonics in the calculation of the frozen eccentricity.

terms are retained in this algorithm, thus small errors of order J_2^2 are to be expected. As it is explained in Appendix G of Schaub and Junkins (2003), the forward and inverse mapping functions between the mean and osculating orbit elements only differ by a sign, because a first-order truncation is performed for the infinite power series solution. In this appendix, the equations are written with the signs corresponding to the change from osculating to mean elements (to compute the osculating elements from the mean ones, it is enough to switch the signs of the short-periodic terms).

Let the original osculating orbit elements be given by $(a, e, i, \Omega, \omega, M)$ and the transformed mean orbit elements be given through $\left(\hat{a}, \hat{e}, \hat{i}, \hat{\Omega}, \hat{\omega}, \widehat{M}\right)$. The true anomaly, v, is computed using Kepler's equation, given in Eq. (69), and the eccentric anomaly, E, is related with v through Eq. (70).

$$M = E - e\sin E, \tag{69}$$

$$v = 2\tan^{-1}\left(\sqrt{\frac{1+e}{1-e}}\tan\left(\frac{E}{2}\right)\right). \tag{70}$$

Following Kozai, the short-periodic term for the orbital elements of the zonal problem are computed by Eqs. (71)–(76) given next. It is important to notice that the mean values of short-periodic perturbations are not zero, except those of semimajor axis. Therefore, in these equations, their mean values with respect to the mean anomaly are subtracted. This subtraction is the last term of Eqs. (72)–(76).

Semimajor axis:

$$a_{sp} = \frac{J_2}{2a}\left[(2-3\kappa)\left(\frac{a^3}{r^3} - \lambda^{-3}\right) + 3\kappa\frac{a^3}{r^3}c_{2,2}\right]. \tag{71}$$

Eccentricity:

$$\begin{aligned} e_{sp} = {} & \tfrac{\lambda^2}{2e}\tfrac{3J_2}{a^2}\left[\tfrac{1}{3}\left(1-\tfrac{3}{2}\kappa\right)\left(\tfrac{a^3}{r^3}-\lambda^{-3}\right)+\tfrac{1}{2}\tfrac{a^3}{r^3}\kappa c_{2,2}\right] \\ & -\tfrac{3J_2\kappa}{4ea^2\lambda^2}\left[c_{2,2}+ec_{1,2}+\tfrac{1}{3}ec_{3,2}\right]-\tfrac{J_2\kappa e(2\lambda+1)\cos(2\omega)}{4a^2\lambda^2(\lambda+1)^2}. \end{aligned} \tag{72}$$

Inclination:

$$\begin{aligned} i_{sp} = {} & \frac{J_2}{8a^2\lambda^4}\sin 2i(3c_{2,2}+3ec_{1,2}+ec_{3,2}) \\ & -\frac{J_2\sin 2i\left(2\lambda^2-\lambda-1\right)\cos(2\omega)}{8a^2\lambda^4(\lambda+1)}. \end{aligned} \tag{73}$$

Argument of pericenter:

$$\begin{aligned} \omega_{sp} = {} & \tfrac{3J_2}{2a^2\lambda^4}\left\{\tfrac{4-5\kappa}{2}\left(v-M+es_{1,0}\right)+\tfrac{5\kappa-2}{4}\left(s_{2,2}+es_{1,2}+\tfrac{e}{3}s_{3,2}\right)\right. \\ & +\tfrac{1}{e}\left[\left(\tfrac{2-3\kappa}{2}\right)\left[\left(1-\tfrac{e^2}{4}\right)s_{1,0}+\tfrac{e}{2}s_{2,0}+\tfrac{e^2}{12}s_{3,0}\right]\right. \\ & \left.\left.-\kappa\left(\tfrac{1}{4}\left(1+\tfrac{5}{4}e^2\right)s_{1,2}-\tfrac{e^2}{16}s_{1,-2}-\tfrac{7}{12}\left(1-\tfrac{e^2}{28}\right)s_{3,2}-\tfrac{3}{8}es_{4,2}-\tfrac{e^2}{16}s_{5,2}\right)\right]\right\} \\ & -\tfrac{3J_2}{2a^2\lambda^4}\left[\tfrac{\kappa}{8}+\tfrac{(1+2\lambda)\left(2\kappa\lambda^2-\lambda^2-\kappa+1\right)}{6(\lambda+1)^2}\right]\sin(2\omega). \end{aligned} \tag{74}$$

Longitude of the ascending node:

$$\begin{aligned} \Omega_{sp} = {} & -\tfrac{3J_2}{2a^2\lambda^4}\cos i\left[v-M+e\sin v-\tfrac{1}{2}s_{2,2}-\tfrac{1}{2}es_{1,2}-\tfrac{1}{6}es_{3,2}\right] \\ & -\tfrac{J_2\cos i\left(2\lambda^2-\lambda-1\right)\sin(2\omega)}{4a^2\lambda^4(\lambda+1)}. \end{aligned} \tag{75}$$

Mean anomaly:

$$\begin{aligned} eM_{sp} = {} & \tfrac{3J_2}{2a^2\lambda^3}\left\{-\left(1-\tfrac{3}{2}\kappa\right)\left[\left(1-\tfrac{e^2}{4}\right)\sin v+\tfrac{e}{2}\sin 2v+\tfrac{e^2}{12}\sin 3v\right]\right. \\ & +\kappa\left[\tfrac{1}{4}\left(1+\tfrac{5}{4}e^2\right)\sin(v+2\omega)-\tfrac{e^2}{16}\sin(v-2\omega)-\tfrac{7}{12}\left(1-\tfrac{e^2}{28}\right)\sin(3v+2\omega)\right. \\ & \left.\left.-\tfrac{3}{8}e\sin(4v+2\omega)-\tfrac{e^2}{16}\sin(5v+2\omega)\right]\right\}+\tfrac{eJ_2\kappa\left(4\lambda^3-\lambda^2-18\lambda-9\right)\sin 2\omega}{16a^2\lambda^3(\lambda+1)^2}. \end{aligned} \tag{76}$$

The following terms were used in Eqs. (71)–(76):

$$\begin{aligned} & \lambda = \sqrt{1-e^2},\ \kappa = \sin^2 i,\ r = \tfrac{a\left(1-e^2\right)}{1+e\cos v}, \\ & s_{1,0} = \sin v,\ s_{2,0} = \sin 2v,\ s_{3,0} = \sin 3v,\ s_{1,2} = \sin(v+2\omega), \\ & s_{1,-2} = \sin(v-2\omega),\ s_{2,2} = \sin(2v+2\omega),\ s_{3,2} = \sin(3v+2\omega), \\ & s_{4,2} = \sin(4v+2\omega),\ s_{5,2} = \sin(5v+2\omega),\ c_{1,2} = \cos(v+2\omega), \\ & c_{2,2} = \cos(2v+2\omega),\ c_{3,2} = \cos(3v+2\omega). \end{aligned} \tag{77}$$

These expressions, with the exception of the semimajor axis, are well known to contain singularities both near-circular and/or equatorial orbits. In order to obtain a more robust mapping near these conditions, Lyddane's theory is applied. To provide numerically stable expressions for the mean anomaly and eccentricity short-periodic components, ς and $\imath$ are defined next by Eqs. (78) and (79). Similarly, for the right ascension of the ascending node and the inclination, ρ and ϕ are determined by Eqs. (80) and (81).

$$\varsigma = (e-e_{sp})\cos M + eM_{sp}\sin M \approx (e-e_{sp})\cos(M-M_{sp}), \tag{78}$$

$$\imath = (e-e_{sp})\sin M - eM_{sp}\cos M \approx (e-e_{sp})\sin(M-M_{sp}), \tag{79}$$

$$\rho = \left(\sin\frac{i}{2}-\frac{i_{sp}}{2}\cos\frac{i}{2}\right)\cos\Omega+\sin\frac{i}{2}\sin\Omega\Omega_{sp} \approx \sin\left(\frac{i-i_{sp}}{2}\right)\cos\left(\Omega-\Omega_{sp}\right), \tag{80}$$

$$\phi = \left(\sin\frac{i}{2}-\frac{i_{sp}}{2}\cos\frac{i}{2}\right)\sin\Omega+\sin\frac{i}{2}\cos\Omega\Omega_{sp} \approx \sin\left(\frac{i-i_{sp}}{2}\right)\sin\left(\Omega-\Omega_{sp}\right). \tag{81}$$

Once these variables are defined, numerically stable expressions for the five orbital elements can be computed with Eqs. (83)-(87), whereas the mean semimajor axis is computed by Eq. (82).

$$\hat{a} = a - a_{sp}, \tag{82}$$

$$\widehat{M} = \tan^{-1}\left(\frac{\imath}{\varsigma}\right), \tag{83}$$

$$\hat{e} = \sqrt{\imath^2+\varsigma^2}, \tag{84}$$

$$\hat{\Omega} = \tan^{-1}\left(\frac{\phi}{\rho}\right), \tag{85}$$

$$\hat{i} = 2\sin^{-1}\left(\sqrt{\phi^2+\rho^2}\right), \tag{86}$$

$$\hat{\omega} = \left(M-M_{sp}\right)+\left(\omega-\omega_{sp}\right)+\left(\Omega-\Omega_{sp}\right)-\widehat{M}-\hat{\Omega}. \tag{87}$$

References

Adushkin, V., Aksenov, O.Y., Veniaminov, S., et al., 2020. The small orbital debris population and its impact on space activities and ecological safety. Acta Astronaut. 176, 591–597. https://doi.org/10.1016/j.actaastro.2020.01.015.

Alarcón Rodríguez, J., Martínez Fadrique, F., & Klinkrad, H. (2002). Collision risk assessment with a "smart sieve" method. In Joint ESA-NASA Space-Flight Safety Conference (p. 159). volume 486.

Alfano, S., Finkleman, D., 2014. On selecting satellite conjunction filter parameters. Acta Astronaut. 99, 193–200. https://doi.org/10.1016/j.actaastro.2014.02.004.

Algethamie, R., Armellin, R., 2020. Probability of collision for a newly generated debris cloud using subset simulation technique. AIP Conf. Proc. 2226, 050004. https://doi.org/10.1063/5.0003216.

Bateman, H., 1938. Halley's methods for solving equations. Am. Math. Month. 45 (1), 11–17.

Bombardelli, C., Falco, G., Amato, D., et al., 2021. Space occupancy in low-earth orbit. J. Guid., Control, Dynam. 44 (4), 684–700. https://doi.org/10.2514/1.G005371.

Casanova, D., Tardioli, C., Lemaître, A., 2014. Space debris collision avoidance using a three-filter sequence. Month. Notices Roy. Astron. Soc. 442 (4), 3235–3242. https://doi.org/10.1093/mnras/stu1065.

Cook, G., 1966. Perturbations of near-circular orbits by the earth's gravitational potential. Planet. Space Sci. 14 (5), 433–444. https://doi.org/10.1016/0032-0633(66)90015-8.

Escobar, D., Águeda, A., Martín, L., et al., 2012. Efficient all vs. all collision risk analyses. J. Aerospace Eng., Sci. Appl. 4 (2), 40–48. https://doi.org/10.7446/jaesa.0402.04.

Fawcett, T., 2006. An introduction to roc analysis. Pattern Recogn. Lett. 27 (8), 861–874. https://doi.org/10.1016/j.patrec.2005.10.010, ROC Analysis in Pattern Recognition.

Hall, R., Berry, M., Coppola, V., et al., 2009. All-on-all conjunction assessment: Methods for automating and minimizing the computation time. In: Ryan, S. (Ed.), Advanced Maui Optical and Space Surveillance Technologies Conference, p. E71.

Healy, L.M., 1995. Close conjunction detection on parallel computer. J. Guid., Control, Dynam. 18 (4), 824–829. https://doi.org/10.2514/3.21465.

Holzinger, M., Scheeres, D., 2009. Reachability analysis applied to space situational awareness. In 2009 Advanced Maui Optical and Space Surveillance Technologies Conference. Maui, HI, pp. 1–4.

Hoots, F.R., Crawford, L.L., Roehrich, R.L., 1984. An analytic method to determine future close approaches between satellites. Celest. Mech. 33, 143–158. https://doi.org/10.1007/BF01234152.

Karrenberg, H.K., 1964. Eclipse fractions for sun-synchronous orbits. Ph. D. Thesis, URL: https://apps.dtic.mil/sti/citations/AD0603152.

Kerr, E., Sanchez, N., 2021. State of the art and future needs in conjunction analysis methods, processes and software. In: Proceedings of 8th European Conference on Space Debris, pp. 20–23.

Kozai, Y., 1959. The motion of a close earth satellite. Astronom. J. 64, 367–377. https://doi.org/10.1086/107957.

Krag, H., Flohrer, T., Merz, K., et al., 2011. Requirements for a leo surveillance radar in response to collision avoidance user needs. EMR 2 (5.0), 5.

Krage, F.J., 2020. Nasa spacecraft conjunction assessment and collision avoidance best practices handbook, Technical report, NASA. URL: https://nodis3.gsfc.nasa.gov/OCE_docs/OCE_50.pdf.

Lyddane, R.H., 1963. Small eccentricities or inclinations in the brouwer theory of the artificial satellite. Astronom. J. 68 (8), 555–558. https://doi.org/10.1086/109179.

Muelhaupt, T.J., Sorge, M.E., Morin, J., et al., 2019. Space trafic management in the new space era. J. Space Saf. Eng. 6 (2), 80–87. https://doi.org/10.1016/j.jsse.2019.05.007.

Pardini, C., Anselmo, L., 2014. Review of past on-orbit collisions among cataloged objects and examination of the catastrophic fragmentation concept. Acta Astronaut. 100, 30–39. https://doi.org/10.1016/j.actaastro.2014.03.013.

Polyak, B.T., 2007. Newton's method and its use in optimization. Eur. J. Oper. Res. 181 (3), 1086–1096.

Rivero, A.S., Bombardelli, C., Vazquez, R., 2022. Fast orbit propagation for conjunction screening. In: Proceedings of the KePASSA 2022. La Rioja, Spain, pp. 22–24.

Rivero, A.S., Bombardelli, C., Vazquez, R., 2023. Space occupancy conjunction filter. In: Proceding of the 33rd AAS/AIAA Space Flight Mechanics Meeting.

Sanchez, J.C., Louembet, C., Gavilan, F., et al., 2021. Event-based impulsive control for spacecraft rendezvous hovering phases. J. Guid., Control, Dynam. 44 (10), 1794–1810.

Sánchez-Ortiz, N., Domínguez-González, R., Krag, H., et al., 2015. Impact on mission design due to collision avoidance operations based on tle or csm information. Acta Astronaut. 116, 368–381.

Schaub, H., Junkins, J.L., 2003. Analytical Mechanics of Aerospace Systems. American Institute of Aeronautics and Astronautics. https://doi.org/10.2514/4.861550.

Sgobba, T., Allahdadi, F.A., 2013. Orbital operations safety. In: Safety Design for Space Operations. Elsevier, pp. 411–602. https://doi.org/10.1016/b978-0-08-096921-3.00008-8.

Stevenson, E., Rodriguez-Fernandez, V., Urrutxua, H., et al., 2023. Benchmarking deep learning approaches for all-vs-all conjunction screening. Adv. Space Res. https://doi.org/10.1016/j.asr.2023.01.036.

Strong, T., 1859. Elementary and higher algebra. Pratt and Oakley, p. 469.

Vallado, D.A., 2001. Fundamentals of astrodynamics and applications, volume 12. Springer Science & Business Media.

Vallado, D.A., Crawford, P., Hujsak, R., et al., 2006. Revisiting spacetrack report# 3: rev 1. In AIAA/AAS Astrodynamics Specialist Conference and Exhibit. https://doi.org/10.2514/6.2006-6753.

Woodburn, J., Coppola, V., Stoner, F., 2009. A description of filters for minimizing the time required for orbital conjunction computations. In: AAS/AIAA Astrodynamics Specialist Conference, Pittsburgh, PA.

Yaglioglu, B., 2012. Mission operations to improve space mission protection. SpaceOps 2012, (p. 1275683).